

Chapter 5

Introduction to the Minimal Model Program

Revised chapter corresponding to original pp. 268–330.

Original numbering and proof order retained.

The accompanying correction report gives the original page locations.

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5.1 Introduction to Mori's program

We work over $\mathbb{C}$ in this section. Recall that in Chapter 2, we proved the result for contracting rational curves on a smooth projective surface:

Theorem 5.1.1 (MMP for Surfaces). *Let X be a smooth projective surface and $R \subseteq \overline{\text{NE}}(X)$ an extremal ray such that $(R \cdot K_X) < 0$, i.e. R is generated by a K_X -negative curve. Then the contraction morphism $\text{cont}_R : X \rightarrow Z$ exists and is one of the following types:*

- (1) Z is a smooth surface and X is obtained from Z by blowing up a closed point; $\rho(Z) = \rho(X) - 1$.
- (2) Z is a smooth curve and X is a $\mathbb{P}^1$ -bundle over Z ; $\rho(X) = 2$.
- (3) Z is a point, $\rho(X) = 1$ and $-K_X$ is ample. (In fact, $X \cong \mathbb{P}^2$, but this is harder to prove.)

The cases 5.1.1(2) and 5.1.1(3) provide a structure theorem for X , and we call these two contractions Mori fibre spaces. Case 5.1.1(1) introduces a new surface Z . We can apply Theorem 5.1.1 to Z and continue if possible. At each birational step the Picard number drops by one, and so this process terminates. This gives the following:

Theorem 5.1.2. *Let X be a smooth projective surface. There is a sequence of birational contractions $X \rightarrow X_1 \rightarrow \cdots \rightarrow X_n = X'$ such that X' satisfies exactly one of the following conditions:*

- (1) $K_{X'}$ is nef;
- (2) X' is a $\mathbb{P}^1$ -bundle over a curve C ;
- (3) $X' \cong \mathbb{P}^2$.

Definition 5.1.3. If $K_{X'}$ is nef as above, then X' is called a minimal model of X .

Remark 5.1.4. When X has non-negative Kodaira dimension, $K_{X'}$ being nef is equivalent to X' being a minimal surface. Without this assumption, $K_{X'}$ being nef still implies that X' is a minimal surface with no (-1) -curves, but the reverse direction fails; see Figure 2.1 for minimal surfaces which are not minimal models in this sense. Therefore, the concepts of minimal surface and minimal model are different. The terminology “minimal model” in the surface theory of [Har77] uses the former convention, whereas Definition 5.1.3 requires nefness of the canonical divisor.

It turns out that, when a minimal model exists, the morphism $X \rightarrow X'$ is unique up to a unique isomorphism of the targets compatible with these morphisms. Thus X' is determined by the surface X . Indeed, suppose that there are two minimal models X_1 and X_2 such that K_{X_1} and K_{X_2} are nef. Via the diagram

$$\begin{array}{ccc} & X & \\ \swarrow & & \searrow \\ X_1 & \overset{\varphi}{\dashrightarrow} & X_2 \end{array}$$

we obtain a rational map $\varphi : X_1 \dashrightarrow X_2$ and eliminate the indeterminacy to find a smooth common resolution

$$\begin{array}{ccc} & W & \\ \swarrow \varphi_1 & & \searrow \varphi_2 \\ X_1 & & X_2 \end{array}$$

Choose such a W with minimal Picard number. If φ_1 is not an isomorphism, there is a (-1) -curve E contracted by φ_1 . It cannot also be contracted by φ_2 : otherwise both morphisms factor through the contraction of E , contradicting the choice of W . Write $C = (\varphi_2)_*E$ on X_2 . Then, by

$$K_W = \varphi_2^*K_{X_2} + \sum_i a_i E_i, \quad a_i > 0,$$

where the E_i are the φ_2 -exceptional curves, we have

$$(K_W \cdot E) = (K_{X_2} \cdot C) + \sum_i a_i (E_i \cdot E).$$

The left hand side is -1 , and the right hand side is non-negative since K_{X_2} is nef and E is distinct from all the E_i . This is a contradiction. Similarly, φ_2 must be an isomorphism.

Now, run the minimal model program on a given smooth projective surface X :

Step 0: We start with a smooth projective surface X .

Step 1: If K_X is nef, then we go to Step 5. If K_X is not nef, then by the Cone Theorem we can choose a K_X -negative extremal ray $R \subseteq \overline{\text{NE}}(X)$.

Step 2: By Theorem 5.1.1, the contraction morphism $\text{cont}_R : X \rightarrow Z$ exists. There are two possibilities.

Step 3: If $\dim Z = \dim X$, then go back to Step 0 with Z replacing X .

Step 4: If $\dim Z < 2 = \dim X$, then 5.1.1(2)–(3) gives a structure theorem for X .

Step 5: If K_X is nef, then we stop. These surfaces should be investigated by other methods.

The aim of the Minimal Model Program is to provide a similar step by step approach to higher dimensional varieties. The next step is to develop an analogue of Theorem 5.1.1 for higher dimensional varieties. There are already numerous cases for smooth threefolds over $\mathbb{C}$; see [Mor82, §3].

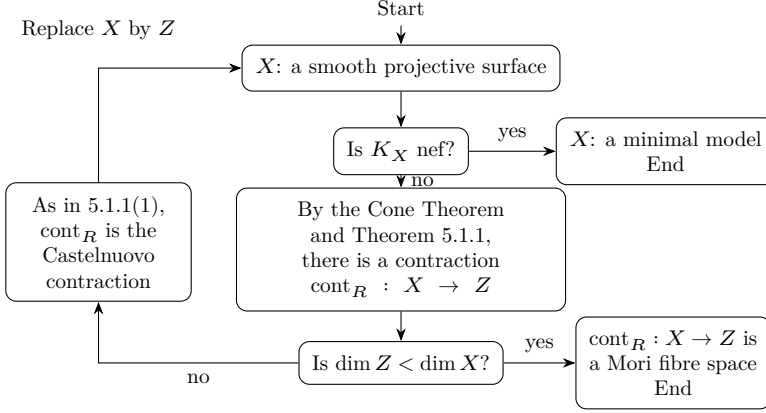


Figure 5.1: Minimal Model Program for Smooth Surfaces.

Theorem 5.1.5 ([Mor82]). *Let X be a non-singular projective threefold over $\mathbb{C}$ and $\text{cont}_R : X \rightarrow Y$ the contraction of a K_X -negative extremal ray $R \subseteq \overline{\text{NE}}(X)$. The following is a list of all possibilities for cont_R :*

E: (Exceptional) $\dim Y = 3$, cont_R is birational, and there are five types of local behaviour near the contracted surface:

E1: cont_R is the blow-up of a smooth curve in the smooth threefold Y .

E2: cont_R is the blow-up of a smooth point of the smooth threefold Y .

E3: cont_R is the blow-up of an ordinary double point of Y . Locally analytically, an ordinary double point is given by $x^2 + y^2 + z^2 + w^2 = 0$.

E4: cont_R is the blow-up of a point of Y which is locally analytically given by $x^2 + y^2 + z^2 + w^3 = 0$.

E5: cont_R contracts a smooth $\mathbb{P}_{\mathbb{C}}^2$ with normal bundle $\mathcal{O}(-2)$ to a point of multiplicity 4 on Y which is locally analytically the quotient of $\mathbb{C}^3$ by the involution $(x, y, z) \mapsto (-x, -y, -z)$.

C: (Conic bundle) $\dim Y = 2$, Y is smooth, and cont_R is a fibration whose fibres are plane conics. General fibres are, of course, smooth.

D: (Del Pezzo fibration) $\dim Y = 1$, Y is a smooth curve, and general fibres of cont_R are Del Pezzo surfaces.

F: (Fano variety) $\dim Y = 0$, $-K_X$ is ample, and hence X is a Fano variety with $\rho(X) = 1$.

Note that because of the singularities occurring in cases E3, E4 and E5, we cannot apply Theorem 5.1.5 to the resulting variety Y . Thus we need to go back and see if one can prove a version of Theorem 5.1.5 when we allow X to have

certain singularities. It turns out that the singularities in E3 and E4 do not cause much trouble. But the singularity in E5 is not a hypersurface singularity, and it is much harder to give a direct geometric analysis of varieties with such singularities.

So it seems that a more conceptual approach is required. As with many questions, there are two opposing points of view:

- **Minimalist:** We should try to identify the smallest class of singularities which we encounter starting with smooth varieties and applying Theorem 5.1.5 and its generalizations.
- **Maximalist:** We should try to identify the largest class of singularities where an analogue of Theorem 5.1.1 is possible.

For the beginning of the section we try to follow the maximalist approach. This is most suitable for general discussion. Unfortunately, essentially none of the results needed for the entire program hold without restrictions on the singularities. Thus, when it comes to proofs, we have to get close to the minimalist approach.

In Step 1 we first ask if K_X is nef. For this question to make sense, the intersection numbers of K_X with curves have to be defined. These numbers make sense if K_X is Cartier, or more generally, when K_X is $\mathbb{Q}$ -Cartier. We retain this assumption in the following discussion.

Next we have to establish the Cone Theorem for X . For smooth projective varieties this was done in Chapter 2. The main use of smoothness is in the bend-and-break method, to get the lower bound for the dimension of the deformation space; recall the deformation estimates in §2.1. This lower bound frequently fails when X is singular. There are, however, large classes of singularities where the estimate holds, and this leads to the Cone Theorem for many singular varieties. In dimension three a Cone Theorem holds for normal projective varieties with K_X $\mathbb{Q}$ -Cartier, whereas the corresponding assertion fails for arbitrary normal varieties in higher dimensions; see [KM98, §2.3] and [Kol92]. This does not by itself give the contraction theorem for arbitrary singularities.

In Step 2 we need to contract the extremal ray $R \subseteq \overline{NE}(X)$. This is a more delicate result, which holds only under suitable assumptions on the singularities. For the moment we recall a result proved in Chapter 2, which classifies the resulting extremal contractions into three types.

Lemma 5.1.6. *Let X be a normal projective variety and let $R \subseteq \overline{NE}(X)$ be an extremal ray. Assume that X is $\mathbb{Q}$ -factorial and that its contraction to a normal projective variety exists:*

$$f := \text{cont}_R : X \longrightarrow Z.$$

Then one of the following holds:

- *f is birational and $\text{Ex}(f)$ is an irreducible divisor; we say that f is a divisorial contraction;*

- f is birational and $\text{Ex}(f)$ has codimension at least two; we say that f is a small contraction;
- $\dim Z < \dim X$; we say that f is of fibre type.

The three cases of Lemma 5.1.6 behave very differently from the point of view of our program. Write the target as Y in the following discussion.

Case 1. Fibre type contractions

These correspond to Step 4. If $(R \cdot K_X) < 0$, then $-K_X$ is relatively ample, since the relative cone is generated by R . The general fibre F is normal, and $-K_F = -K_X|_F$ is ample. We call this a Fano fibre space. Thus, at least in principle, we reduce the problem of understanding X to the study of the lower dimensional variety Y and the fibres of f . Moreover, the general fibres are of a very special kind: they are analogues of $\mathbb{P}_{\mathbb{C}}^1$ and of Del Pezzo surfaces.

Case 2. Divisorial contractions

These should correspond to Step 3. For this to be true, we need to establish that K_Y is $\mathbb{Q}$ -Cartier, and preferably that Y is $\mathbb{Q}$ -factorial. If this holds, then Y can be considered simpler than X , as the Picard number goes down by one.

Case 3. Small contractions

This is a new situation which never happens for surfaces for dimensional reasons. It also does not occur among the K_X -negative extremal contractions of smooth threefolds in Theorem 5.1.5. Small K_X -trivial contractions of smooth threefolds can occur, as the auxiliary construction in Example 5.1.7 will show.

For a small K_X -negative contraction, we claim that Y has singularities for which no positive multiple of K_Y is Cartier. Indeed, assume that mK_Y and mK_X are Cartier for some $m > 0$. Choose compatible canonical divisors. Then mK_X and $f^*(mK_Y)$ agree outside $\text{Ex}(f)$, since f is an isomorphism there. Their difference is a Weil divisor supported in codimension at least two, so it is zero. Hence

$$mK_X = f^*(mK_Y).$$

This is impossible: $(R \cdot K_X) < 0$, whereas $(R \cdot f^*(mK_Y)) = 0$, because the curves in R are contracted to points.

So we have been led out of the class of varieties that we can control. In order to continue at this point, we introduce a new operation called a flip. The flip produces a new variety X^+ and a canonical divisor K_{X^+} which is positive on the new contracted ray R^+ . In this sense, it makes a step toward the goal of obtaining a minimal model, where the canonical divisor is nef.

Example 5.1.7 (Example of flips). In this example, for $n \geq 2$, the flip removes the curve $C_n \cong \mathbb{P}^1$ from the singular variety X_n and replaces it with $C_n^+ \cong \mathbb{P}^1$ to obtain the variety X_n^+ , which in this case is non-singular.

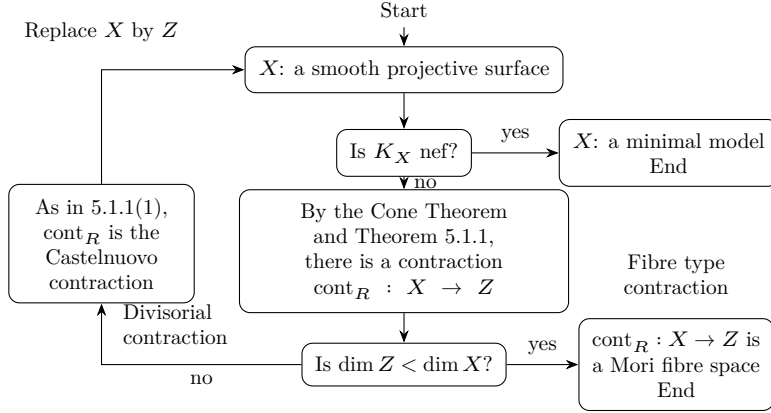


Figure 5.2: Two possibilities correspond to different contractions.

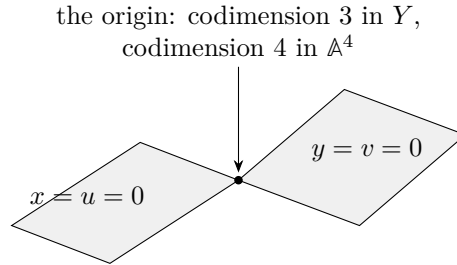


Figure 5.3: Informal diagram of the two planes through the origin.

We start with an auxiliary construction. Consider $Y = V(xy - uv) \subseteq \mathbb{A}^4$, which has an isolated singularity at the origin.

To investigate $\widetilde{X} = \text{Bl}_0 Y$, we blow up $\mathbb{A}^4$ at the origin:

$$\text{Bl}_0 \mathbb{A}^4 = \left\{ ((x, y, u, v), [X : Y : U : V]) \in \mathbb{A}^4 \times \mathbb{P}^3 : \text{rank} \begin{pmatrix} x & y & u & v \\ X & Y & U & V \end{pmatrix} = 1 \right\}.$$

The projection $\pi : \text{Bl}_0 \mathbb{A}^4 \rightarrow \mathbb{A}^4$ is given by the affine coordinates. Thus we have the equations

$$\begin{cases} xY = yX, & xU = uX, & xV = vX, \\ yU = uY, & yV = vY, & uV = vU. \end{cases}$$

At the origin, the preimage is $\{0\} \times \mathbb{P}^3 \cong \mathbb{P}^3$, with no relation among X, Y, U, V . Now $\mathbb{P}^3$ is covered by four affine pieces:

- On $X \neq 0$, set $X = 1$. The equations of the total transform of Y are

$$xU = u, \quad xY = y, \quad xV = v, \quad xy - uv = 0.$$

Substituting into the last equation gives $x^2(Y - UV) = 0$. Here $x = 0$ defines the exceptional divisor E of $\text{Bl}_0 \mathbb{A}^4$, the factor x^2 records its multiplicity two in the total transform, and $Y - UV = 0$ defines the strict transform $\widetilde{X}$. The exceptional set Q of $\widetilde{X}$ is $E \cap \widetilde{X}$.

- On $Y \neq 0$, set $Y = 1$. We have

$$x = yX, \quad u = yU, \quad v = yV, \quad xy - uv = 0.$$

Substituting gives $y^2(X - UV) = 0$, where $y = 0$ defines E and $X - UV = 0$ defines $\widetilde{X}$.

Similarly, we obtain the charts for $U \neq 0$ and $V \neq 0$. Therefore the strict transform is

$$\begin{aligned} \widetilde{X} &\subseteq \mathbb{A}^4 \times \mathbb{P}^3, & XY - UV &= 0, \\ \text{rank} \begin{pmatrix} x & y & u & v \\ X & Y & U & V \end{pmatrix} &= 1. \end{aligned}$$

Its fibre over the origin is $Q = V(XY - UV) \subseteq \mathbb{P}^3$. This is the quadric surface given by the Segre embedding $\mathbb{P}^1 \times \mathbb{P}^1 \hookrightarrow \mathbb{P}^3$, so Q has two families of lines:

$$\begin{cases} tX = sV, \\ sY = tU, \end{cases} \quad \text{and} \quad \begin{cases} tX = sU, \\ sY = tV, \end{cases} \quad [s : t] \in \mathbb{P}^1.$$

For $st \neq 0$, writing $c = s/t$, these become the original expressions

$$\begin{cases} X = cV, \\ Y = c^{-1}U, \end{cases} \quad \text{and} \quad \begin{cases} X = cU, \\ Y = c^{-1}V. \end{cases}$$

Indeed, $\det \begin{pmatrix} X & Y \\ U & V \end{pmatrix} = 0$; the first family fixes the proportionality of the columns, and the second that of the rows. The homogeneous form also includes the lines for $c = 0, \infty$.

These two families can be blown down to smooth threefolds X and X^+ . They can also be obtained by blowing up the ideals (x, v) and (x, u) on Y . In detail:

- Blowing up $(xy - uv = 0)$ along the plane $(x = v = 0)$ gives

$$\begin{aligned} X = \text{Bl}_{(x,v)} Y &= \text{Proj}_Y \bigoplus_{d \geq 0} (x, v)^d \\ &\cong \text{Proj}_Y (A[S, T] / \ker \varphi) \\ &= \{((x, y, u, v), [S : T]) : xT = vS, yS = uT\} \\ &\subseteq \mathbb{A}^4 \times \mathbb{P}^1_{[S:T]}, \end{aligned}$$

where $A = \mathbb{C}[x, y, u, v] / (xy - uv)$ and the graded surjection

$$\varphi : A[S, T] \longrightarrow \bigoplus_{d \geq 0} (x, v)^d, \quad S \longmapsto x, \quad T \longmapsto v$$

has kernel $(xT - vS, yS - uT)$. Here x, v on the right are in degree one. Equivalently, the defining ideal in $\mathbb{C}[x, y, u, v, S, T]$ consists of the 2×2 minors of

$$\begin{pmatrix} x & u & S \\ v & y & T \end{pmatrix}.$$

We may omit $xy - uv = 0$ from the incidence equations in $\mathbb{A}^4 \times \mathbb{P}^1$: the other two equations give

$$T(xy - uv) = 0, \quad S(xy - uv) = 0,$$

and S, T cannot vanish simultaneously.

- Blowing up $(xy - uv = 0)$ along the plane $(x = u = 0)$ gives

$$\begin{aligned} X^+ = \text{Bl}_{(x,u)} Y &= \text{Proj}_Y \bigoplus_{d \geq 0} (x, u)^d \\ &\cong \text{Proj}_Y (A[S', T'] / \ker \varphi') \\ &= \{((x, y, u, v), [S' : T']) : xT' = uS', yS' = vT'\} \\ &\subseteq \mathbb{A}^4 \times \mathbb{P}^1_{[S':T']}, \end{aligned}$$

where $\ker \varphi' = (xT' - uS', yS' - vT')$ for the analogous surjection

$$\varphi' : A[S', T'] \longrightarrow \bigoplus_{d \geq 0} (x, u)^d, \quad S' \longmapsto x, \quad T' \longmapsto u.$$

Now consider the two rulings:

- (1) For $X = cV$, $Y = c^{-1}U$, we have $X/V = U/Y = c$ wherever the ratios are defined. Thus

$$\widetilde{X} \longrightarrow X \longrightarrow Y, \quad Q \longrightarrow C \longrightarrow \{0\}.$$

For $X = V(xT - vS, yS - uT) \subseteq \mathbb{A}^4 \times \mathbb{P}^1$, we have

$$\begin{aligned} \text{on } \{S \neq 0\} : \quad T/S &= v/x = y/u = c^{-1}, \\ \text{on } \{T \neq 0\} : \quad S/T &= x/v = u/y = c. \end{aligned}$$

The ratios on Q are interpreted using $[X : Y : U : V]$ and extend across the vanishing denominators through the incidence equations. They determine the point $[S : T]$ of $C \cong \mathbb{P}^1$.

$$\begin{array}{ccc} \mathbb{P}^1 \times \mathbb{P}^1 \cong Q & & \\ \begin{array}{|c|c|c|c|c|} \hline & & & & \\ \hline & & & & \\ \hline & & & & \\ \hline & & & & \\ \hline \end{array} & \xrightarrow{\text{blow down}} & \left| \begin{array}{l} [S : T] \in C \cong \mathbb{P}^1 \end{array} \right. \end{array}$$

- (2) For $X = cU$, $Y = c^{-1}V$, we have $X/U = V/Y = c$. Thus

$$\widetilde{X} \longrightarrow X^+ \longrightarrow Y, \quad Q \longrightarrow C^+ \longrightarrow \{0\}.$$

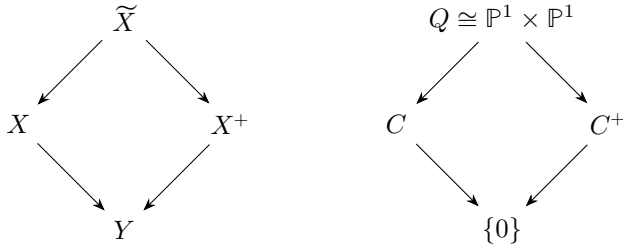
For $X^+ = V(xT' - uS', yS' - vT')$, the corresponding ratios are

$$\begin{aligned} \text{on } \{S' \neq 0\} : \quad T'/S' &= u/x = y/v = c^{-1}, \\ \text{on } \{T' \neq 0\} : \quad S'/T' &= x/u = v/y = c. \end{aligned}$$

They determine a point $[S' : T']$ of $C^+ \cong \mathbb{P}^1$ in the same way.

$$\begin{array}{ccc} \mathbb{P}^1 \times \mathbb{P}^1 \cong Q & & \\ \begin{array}{|c|c|c|c|c|} \hline & & & & \\ \hline & & & & \\ \hline & & & & \\ \hline & & & & \\ \hline \end{array} & \xrightarrow{\text{blow down}} & \begin{array}{c} \text{---} \\ [S' : T'] \in C^+ \cong \mathbb{P}^1 \end{array} \end{array}$$

Therefore we have the following diagram, together with its exceptional loci:



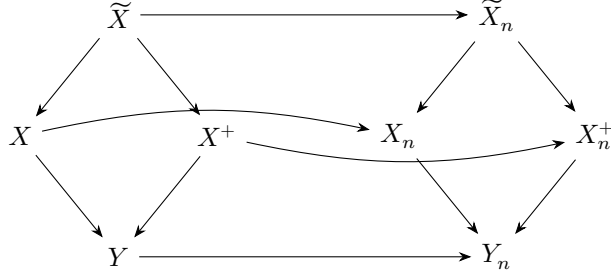
Now consider the action of the cyclic group μ_n , where $n \geq 2$ and $\zeta = \exp(2\pi i/n)$:

$$(x, y, u, v) \mapsto (\zeta x, y, \zeta u, v).$$

This lifts to $\widetilde{X}, X, X^+$ by the actions

$$\begin{aligned} [X : Y : U : V] &\mapsto [\zeta X : Y : \zeta U : V], \\ [S : T] &\mapsto [\zeta S : T], \quad [S' : T'] \mapsto [S' : T']. \end{aligned}$$

We denote the quotients by $\widetilde{X}_n, X_n, X_n^+$ and Y_n . The natural morphisms are equivariant, giving the diagram



On Y , the fixed point set is the two-plane $(x = u = 0)$. On the quadric $Q \subseteq \widetilde{X}$, the projective action has two fixed lines:

- $X = U = 0$, on which $[\zeta X : Y : \zeta U : V] = [X : Y : U : V]$;
- $Y = V = 0$, on which $[\zeta X : Y : \zeta U : V] = [\zeta X : \zeta Y : \zeta U : \zeta V]$.

On X , the fixed locus has two components:

- the strict transform of the plane $(x = u = 0) \subseteq Y$, a surface meeting C at one point;
- the image of the line $(Y = V = 0) \subseteq Q$, an isolated fixed point. On the chart $S = 1$, local coordinates are $(x, v' = v/x, u)$, and the action is

$$(x, v', u) \mapsto (\zeta x, \zeta^{-1} v', \zeta u).$$

On X^+ , the blow-up relation is $xT' = uS'$. On $S' = 1$, local coordinates are $(x, u' = u/x, v)$, and the action is

$$(x, u', v) \mapsto (\zeta x, u', v).$$

The fixed locus here is $x = 0$. On $T' = 1$, the coordinates are $(u, x' = x/u, y)$, with action $(u, x', y) \mapsto (\zeta u, x', y)$, and the fixed locus is $u = 0$. These pieces glue to an irreducible surface containing C^+ . Moreover, X_n^+ is smooth: its two charts are

$$\text{Spec } \mathbb{C}[x^n, u', v], \quad \text{Spec } \mathbb{C}[u^n, x', y].$$

Now we compute the intersections of the canonical classes with the exceptional curves. At the isolated quotient singularity of X_n , the weights are $(1, -1, 1)$. The action is free away from the origin of this chart, and its character on a top form is ζ . Consequently K_{X_n} has index n there. On the other chart the quotient is smooth, so nK_{X_n} is Cartier everywhere.

On $S = 1$, the curve C is the v' -axis and the volume form

$$\omega_1 = dx \wedge dv' \wedge du$$

is sent to $\zeta\omega_1$. Thus

$$\omega_1^{\otimes n} \mapsto \zeta^n \omega_1^{\otimes n} = \omega_1^{\otimes n}.$$

It descends to a non-vanishing local section of $\mathcal{O}_{X_n}(nK_{X_n})$ on this quotient chart. On $T = 1$, use coordinates $(y, u' = u/y, v)$, with C the u' -axis. The form

$$\omega_2 = dy \wedge du' \wedge dv$$

is also sent to $\zeta\omega_2$. Its n th tensor power is invariant, but, because this chart has a divisorial fixed locus, it need only descend as a *rational* pluricanonical form.

More precisely, the quotient coordinates are (y, w, v) , where $w = (u')^n$, and C_n is the w -axis. A regular frame of $\mathcal{O}_{X_n}(nK_{X_n})$ is $(dy \wedge dw \wedge dv)^{\otimes n}$, whose pullback equals

$$(n(u')^{n-1})^n \omega_2^{\otimes n} = n^n (u')^{n(n-1)} \omega_2^{\otimes n}.$$

Thus, as rational forms on the quotient,

$$\omega_2^{\otimes n} = \frac{1}{n^n} w^{-(n-1)} (dy \wedge dw \wedge dv)^{\otimes n}.$$

On the overlap, $v' = v/x = (u/y)^{-1} = (u')^{-1}$ and $x = u'v$, $u = u'y$. Hence

$$\begin{aligned} \omega_1 &= d(u'v) \wedge d((u')^{-1}) \wedge d(u'y) \\ &= -dv \wedge du' \wedge dy = dy \wedge du' \wedge dv = \omega_2. \end{aligned}$$

Gluing these expressions, we obtain a rational section of $\mathcal{O}_{X_n}(nK_{X_n})$. Its restriction to C_n is regular and non-zero on the first chart and has a pole of order $n-1$ at $w = 0$ on the second chart. Therefore

$$(K_{X_n} \cdot C_n) = -\frac{n-1}{n}.$$

Similarly, for the smooth variety X_n^+ , on $S' = 1$ use $(x^n, u' = u/x, v)$ with volume form

$$\omega_1 = d(x^n) \wedge du' \wedge dv.$$

On $T' = 1$, use $(u^n, x' = x/u, y)$ with

$$\omega_2 = d(u^n) \wedge dx' \wedge dy.$$

On the overlap, $x^n = u^n(x')^n$, $u' = (x')^{-1}$ and $v = x'y$, so

$$\omega_1 = -(x')^{n-1}\omega_2.$$

The coordinates u', x' on C^+ are invariant under μ_n , and so also give coordinates on C_n^+ . The transition formula therefore shows that

$$(K_{X_n^+} \cdot C_n^+) = n - 1.$$

The quotient morphisms $X_n \rightarrow Y_n$ and $X_n^+ \rightarrow Y_n$ are projective and are isomorphisms away from these exceptional curves. Their relative cones are each generated by the corresponding curve. The two intersection computations imply that $-K_{X_n}$ is relatively ample and $K_{X_n^+}$ is relatively ample. Thus $X_n^+ \rightarrow Y_n$ is the flip of $X_n \rightarrow Y_n$ for $n \geq 2$. For $n = 1$, the auxiliary construction has zero canonical degree on both exceptional curves and gives the Atiyah flop.

The operation in the above example can be formalized as follows:

Definition 5.1.8 (Flip). Let $f : X \rightarrow Y$ be a proper birational morphism of normal varieties such that the exceptional set $\text{Ex}(f)$ has codimension at least two in X . Assume furthermore that K_X is $\mathbb{Q}$ -Cartier and $-K_X$ is f -ample. A normal variety X^+ together with a proper birational morphism $f^+ : X^+ \rightarrow Y$ is called a flip of f if

- (1) K_{X^+} is $\mathbb{Q}$ -Cartier;
- (2) K_{X^+} is f^+ -ample;
- (3) $\text{Ex}(f^+)$ has codimension at least two in X^+ .

By a slight abuse of terminology, the rational map $\varphi : X \dashrightarrow X^+$ is also called a flip.

We will see that a flip is unique over Y , and the main question is its existence. A flip gives the following diagram:

$$\begin{array}{ccc}
 X & \xrightarrow{\quad \varphi \quad} & X^+ \\
 \searrow f & & \swarrow f^+ \\
 & Y & \\
 \text{\scriptsize } -K_X \text{ is } f\text{-ample} & & \text{\scriptsize } K_{X^+} \text{ is } f^+\text{-ample}
 \end{array}$$

If we perform a flip, it is not clear that X^+ is simpler than X . In the example above this happens since X_n is singular but X_n^+ is smooth.

Before we formalize the above discussion, it is necessary to settle on a class of singularities where the Minimal Model Program works. Such a class was already

studied in the nineteenth century; see [Die85, VI, 23], in the following setting. Let $Y \subseteq \mathbb{P}^n$, $n \geq 2$, be a normal hypersurface of degree d , possibly singular, and let $f : X \rightarrow Y$ be a resolution. The adjunction formula gives

$$\omega_Y \cong \mathcal{O}_Y(d - n - 1).$$

One would like to have

$$H^0(X, \mathcal{O}_X(K_X)) \stackrel{?}{=} H^0(Y, \mathcal{O}_Y(K_Y)) = H^0(Y, \mathcal{O}_Y(d - n - 1)).$$

The last space has dimension $\binom{d-1}{n}$ (interpreted as zero if $d - 1 < n$), by the hypersurface exact sequence and the cohomology of $\mathbb{P}^n$. But the first equality can fail when Y is singular, because its singularities impose “adjunction conditions”. Thus it is of interest to understand which singularities impose no adjunction conditions, also for pluricanonical forms. In modern terminology these are canonical singularities:

Definition 5.1.9 (Canonical singularities). Let Y be a normal variety. We say that Y has canonical singularities if

- (1) K_Y is $\mathbb{Q}$ -Cartier;
- (2) for every resolution $f : X \rightarrow Y$ and every positive integer m divisible by the Cartier index of K_Y , the natural inclusion is an equality:

$$f_*\mathcal{O}_X(mK_X) = \mathcal{O}_Y(mK_Y).$$

Both sides are identified inside the sheaf of rational pluricanonical forms on Y .

The following observation leads to a smaller class. If Y is smooth, $m > 0$, and $\gamma \in H^0(Y, \mathcal{O}_Y(mK_Y))$, then $f^*\gamma$, as a section of $\mathcal{O}_X(mK_X)$, vanishes along every exceptional prime divisor. This is stronger than condition (2) above, which requires only that $f^*\gamma$ have no poles there:

$$H^0(X, \mathcal{O}_X(mK_X)) = H^0(Y, f_*\mathcal{O}_X(mK_X)) \cong H^0(Y, \mathcal{O}_Y(mK_Y)), \quad f^*\gamma \longleftrightarrow \gamma.$$

This stronger condition gives the next definition, the class used when running the MMP starting from smooth varieties.

Definition 5.1.10 (Terminal singularities). Let Y be a normal variety. We say that Y has terminal singularities if

- (1) K_Y is $\mathbb{Q}$ -Cartier;
- (2) for every resolution $f : X \rightarrow Y$ and every positive integer m divisible by the Cartier index of K_Y , the natural inclusion is an equality:

$$f_*\mathcal{O}_X(mK_X - E) = \mathcal{O}_Y(mK_Y),$$

where $E \subseteq X$ is the reduced exceptional divisor, i.e. the divisorial part of $\text{Ex}(f)$.

We introduce the concept of higher dimensional minimal models:

Definition 5.1.11 (Minimal model). Let X be a normal proper variety. We say that X is minimal, or is a minimal model, if

- (1) X has terminal singularities;
- (2) K_X is nef.

If Y is a smooth proper variety birational to X , then X is also called a minimal model of Y .

It should be stressed that, unlike for surfaces (see the discussion after Remark 5.1.4), higher dimensional minimal models are not unique. With these definitions at hand, we can formulate a step by step outline of the Minimal Model Program.

Minimal Model Program

Starting with a smooth projective variety X , we perform a sequence of birational modifications until we arrive at a variety X^* whose global structure is simpler. The following is a more detailed list of the necessary steps.

Step 0 (Initial datum): We start with a projective $\mathbb{Q}$ -factorial variety $X = X_0$ over $\mathbb{C}$ with terminal singularities. We set up a recursive procedure which produces intermediate varieties X_i , with the aim of stopping at a final variety X^* . Assume that we have already constructed X_i .

Step 1 (Preparation): If K_{X_i} is nef, then there is nothing to do and we go directly to Step 3(2). If K_{X_i} is not nef, then we use the following two results:

- (1) (Cone Theorem)

$$\overline{\text{NE}}(X_i) = \overline{\text{NE}}(X_i)_{K_{X_i} \geq 0} + \sum_j \mathbb{R}_{\geq 0}[C_j].$$

If K_{X_i} is not nef, we can pick a K_{X_i} -negative extremal ray $R_i \subseteq \overline{\text{NE}}(X_i)$.

- (2) (Contraction of an extremal ray) Let $\text{cont}_{R_i} : X_i \rightarrow Y_i$ denote the corresponding extremal contraction.

Step 2 (Birational transformations): By Lemma 5.1.6, there are three possible types of contractions $\text{cont}_{R_i} : X_i \rightarrow Y_i$. Two of them are used to produce a new variety X_{i+1} :

- (1) (Divisorial contraction) If cont_{R_i} is divisorial, set $X_{i+1} = Y_i$. The preservation theorem says that X_{i+1} is again $\mathbb{Q}$ -factorial with terminal singularities, so we go back to Step 0 with X_{i+1} and start anew.

- (2) (Flipping contraction) If cont_{R_i} is small, set $X_{i+1} = X_i^+$, the flip of cont_{R_i} , provided that it exists. The preservation theorem again gives $\mathbb{Q}$ -factorial terminal singularities, so we return to Step 0 with X_{i+1} .

Step 3 (Final outcome): We hope that eventually the procedure stops, giving one of the following two possibilities:

- (1) (Mori fibre space) If $\text{cont}_{R_i} : X_i \rightarrow Y_i$ is a Fano contraction, or fibre type contraction, set $X^* = X_i$. The hope is that the investigation of the lower dimensional variety Y_i and of the fibres provides new methods to study X^* .
- (2) (Minimal model) If K_{X_i} is nef, again set $X^* = X_i$. We hope to exploit the semi-positivity of the canonical class in further attempts to understand X^* .

Unlike divisorial contractions, flips do not decrease the Picard number. Thus the surface argument does not establish termination of this program in higher dimensions; termination is a separate issue.

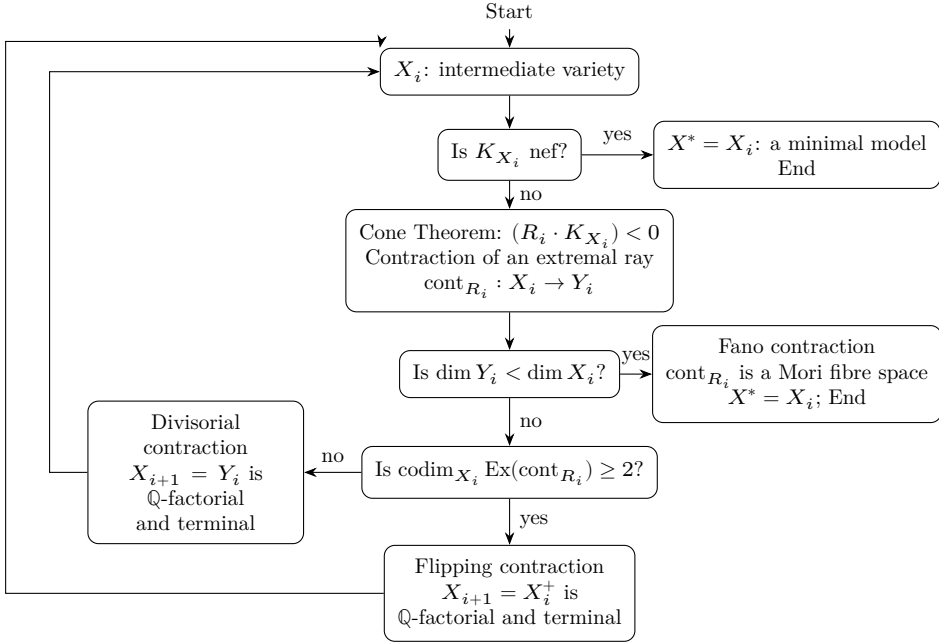


Figure 5.4: Minimal Model Program.

5.2 Singularities in the Minimal Model Program

As we saw in Theorem 5.1.5, contractions of extremal rays lead to singular varieties, as in E3–E5. In order to continue, we have to understand the singularities that occur in the process, and we need a new way to measure how singular a variety is. This new measure is called discrepancy.

Unless otherwise stated, we work over $\mathbb{C}$ and consider varieties of dimension at least two. This dimension convention is needed for the bounds on the infimum over exceptional divisors below. For a normal curve there are no exceptional divisorial valuations over the curve, and the infimum over that empty set would be $+\infty$.

Definition 5.2.1 (Canonical divisor). Let X be a normal variety, and let X_{reg} be its smooth locus. A canonical divisor K_X is a Weil divisor such that

$$\mathcal{O}_X(K_X)|_{X_{\text{reg}}} \cong \omega_{X_{\text{reg}}}.$$

It is well-defined up to linear equivalence.

For a Weil divisor D , the corresponding sheaf $\mathcal{O}_X(D)$ need not be invertible, i.e. a line bundle. It is called a divisorial sheaf and is defined, for an open subset $U \subseteq X$, by

$$H^0(U, \mathcal{O}_X(D)) = \{h \in \mathbb{C}(X)^\times : \text{div}(h)|_U + D|_U \geq 0\} \cup \{0\}.$$

As in the Cartier divisor case in Chapter 1, a non-zero global section s corresponds to a rational function h . Its divisor of zeros is

$$\text{div}(s) = \text{div}(h) + D,$$

which is effective. The same equation defines the divisor of a non-zero rational section, but it need not then be effective. For example, as in Example 1.1.11, the rational section corresponding to $h = 1$ has divisor D .

There is an isomorphism $\mathcal{O}_X(D)_\eta \cong \mathcal{O}_{X,\eta} = \mathbb{C}(X)$ at the generic point, since

$$\mathcal{O}_X(D)_\eta = \varinjlim_{\eta \in U} H^0(U, \mathcal{O}_X(D)),$$

and we can shrink U to avoid the supports of D and $\text{div}(h)$. A morphism $\mathcal{O}_X(D)|_U \rightarrow \mathcal{O}_U$ is therefore generically multiplication by a rational function f :

$$H^0(U, \mathcal{O}_X(D)) \longrightarrow H^0(U, \mathcal{O}_X), \quad h \longmapsto hf.$$

At a prime divisor $P \subseteq U$, write $d_P = \text{coeff}_P D$. Since a generator of $\mathcal{O}_X(D)$ at its generic point has order $-d_P$, this multiplication is regular precisely when

$$v_P(f) - d_P \geq 0.$$

Normality allows us to test this in codimension one. Consequently

$$\mathcal{O}_X(D)^\vee := \mathcal{H}om_{\mathcal{O}_X}(\mathcal{O}_X(D), \mathcal{O}_X) \cong \mathcal{O}_X(-D).$$

Thus $\mathcal{O}_X(D)$ is a reflexive sheaf of rank one: it is isomorphic to its double dual. See also [Sch]. Linearly equivalent divisors have isomorphic divisorial sheaves, though their realizations as subsheaves of $\mathbb{C}(X)$ need not be identical.

5.2A Definition of Discrepancies

Definition 5.2.2 (Discrepancy). Let X be a normal variety such that mK_X is Cartier for some $m > 0$. Suppose that $f : Y \rightarrow X$ is a not necessarily proper birational morphism from a normal variety Y . Let $E \subseteq Y$ be a prime exceptional divisor. Pick a general closed point $e \in E$ where Y and E are smooth, and choose local parameters $y_1, \dots, y_n$ with $E = (y_1 = 0)$. Locally near e ,

$$f^*(\text{local generator of } \mathcal{O}_X(mK_X)) = y_1^{ma(E,X)} u (dy_1 \wedge \dots \wedge dy_n)^{\otimes m},$$

where u is a unit and $ma(E, X) \in \mathbb{Z}$. The rational number $a(E, X)$ is called the discrepancy of E with respect to X .

Proposition 5.2.3. $a(E, X)$ is independent of the choice of m .

Proof. Choose positive integers m, m' such that mK_X and $m'K_X$ are Cartier. Then $mm'K_X$ is Cartier. For local generators ω_X, ω'_X of the corresponding sheaves, we have

$$\begin{aligned} f^*\omega_X &= y_1^{ma(E,X)} u (dy_1 \wedge \dots \wedge dy_n)^{\otimes m}, \\ f^*\omega'_X &= y_1^{m'a'(E,X)} u' (dy_1 \wedge \dots \wedge dy_n)^{\otimes m'}. \end{aligned}$$

The powers $\omega_X^{\otimes m'}$ and $(\omega'_X)^{\otimes m}$ are generators of the same line bundle and differ by a unit. Comparing their pullbacks gives

$$mm'a(E, X) = mm'a'(E, X),$$

so $a(E, X) = a'(E, X)$. □

Proposition 5.2.4. In the above setting, assume that $f : Y \rightarrow X$ is proper and that mK_Y is also Cartier. Then

$$mK_Y \sim_{\text{lin}} f^*(mK_X) + \sum_i ma(E_i, X)E_i,$$

where the E_i are the f -exceptional prime divisors.

Proof. On the locus where f is an isomorphism, the canonical sheaves are naturally identified, so the difference is supported on the exceptional divisors. Write

$$mK_Y \sim_{\text{lin}} f^*(mK_X) + \sum_i c_i E_i.$$

At the generic point of E_i , the local ring of the normal variety Y is a discrete valuation ring, with uniformizer y_1 . To express the differential form, take parameters at a general closed smooth point of E_i as above. Then

$$f^*(\text{local generator of } \mathcal{O}_X(mK_X)) = y_1^{ma(E_i, X)} u(dy_1 \wedge \cdots \wedge dy_n)^{\otimes m}. \quad (5.1)$$

Thus $c_i = ma(E_i, X)$. Equivalently,

$$\mathcal{O}_Y(mK_Y) \cong f^*\mathcal{O}_X(mK_X) \otimes \mathcal{O}_Y\left(\sum_i ma(E_i, X)E_i\right).$$

The sum here is Cartier, although its individual prime components need not be. There is no inclusion in a fixed direction without a sign condition on the coefficients. Choosing compatible canonical divisors gives an equality of Weil divisors, and dividing by m gives

$$K_Y = f^*K_X + \sum_i a(E_i, X)E_i.$$

□

Remark 5.2.5. On a normal variety Y , the one-dimensional local ring $\mathcal{O}_{Y, \eta_E} \subseteq \mathbb{C}(Y)$ at the generic point of a prime divisor E is a discrete valuation ring. It determines a normalized valuation $v(E, Y)$ of $\mathbb{C}(Y) \cong \mathbb{C}(X)$. Such valuations are called algebraic, or divisorial, valuations.

Let $f' : Y' \rightarrow X$ be another birational morphism and $E' \subseteq Y'$ a prime divisor such that $v(E', Y') = v(E, Y)$. This is equivalent to the induced birational map $Y \dashrightarrow Y'$ identifying open neighbourhoods of the generic points of E and E' . The definition then gives $a(E, X) = a(E', X)$.

Thus the discrepancy depends only on the valuation, not on the particular choice of f and Y . This is why these do not appear in the notation $a(E, X)$.

Definition 5.2.6. Let X be a variety, $f : Y \rightarrow X$ a not necessarily proper birational morphism from a normal variety, and $E \subseteq Y$ a prime divisor. Such an E is called a divisor over X . The closure of $f(E)$ in X is called its center on X , denoted by $\text{center}_X E$.

As above, the center depends only on the valuation and its center on X . We now generalize discrepancy to pairs (X, Δ) such that $K_X + \Delta$ is $\mathbb{Q}$ -Cartier.

Definition 5.2.7 (Generalized discrepancy). Let X be a normal variety and $\Delta = \sum_i a_i D_i$ a sum of distinct prime divisors with rational coefficients. Assume that $K_X + \Delta$ is $\mathbb{Q}$ -Cartier. Choose $m > 0$ such that $m(K_X + \Delta)$ is Cartier and $m\Delta$ is integral. Let $f : Y \rightarrow X$ be a birational morphism from a normal variety, and set

$$f_*^{-1}\Delta = \sum_i a_i f_*^{-1}D_i.$$

For an exceptional prime divisor E_i , choose a general closed point $e_i \in E_i$ and parameters as in Definition 5.2.2, avoiding the strict transform of Δ . Locally near e_i ,

$$\begin{aligned} & f^*(\text{local generator of } \mathcal{O}_X(m(K_X + \Delta))) \\ &= y_1^{ma(E_i, X, \Delta)} u(\text{local generator of } \mathcal{O}_Y(m(K_Y + f_*^{-1}\Delta))). \end{aligned}$$

This defines $a(E_i, X, \Delta) \in \mathbb{Q}$, with $ma(E_i, X, \Delta) \in \mathbb{Z}$.

Proposition 5.2.8. *In the above situation, on the locus $Y \setminus E$ where f is a local isomorphism, the following line bundles are naturally isomorphic:*

$$\mathcal{O}_Y(m(K_Y + f_*^{-1}\Delta))|_{Y \setminus E} \cong f^* \mathcal{O}_X(m(K_X + \Delta))|_{Y \setminus E}.$$

Here E denotes the non-isomorphism locus; for a proper birational morphism it is $\text{Ex}(f)$. The sheaf on the left need not be a line bundle on all of Y .

Proof. This follows from the local isomorphism and the identification of canonical forms there. $\square$

As before, it follows that

$$m(K_Y + f_*^{-1}\Delta) \sim_{\text{lin}} f^*(m(K_X + \Delta)) + \sum_{E_i \text{ exceptional}} ma(E_i, X, \Delta) E_i.$$

We call $a(E, X, \Delta)$ the discrepancy of E with respect to the pair, and write $a(E)$ when no confusion is likely. Twisting by $\mathcal{O}_X(m\Delta)$ allows poles along $m\Delta$; consequently we extend the definition to divisors on X by

$$a(D_i, X, \Delta) = -a_i,$$

and $a(D, X, \Delta) = 0$ for any prime divisor D different from the D_i . As in the case $\Delta = 0$, discrepancy depends on the valuation and not on f .

Remark 5.2.9. Dividing the preceding relation by m gives a $\mathbb{Q}$ -linear equivalence of Weil divisors:

$$\begin{aligned} K_Y + f_*^{-1}\Delta &\sim_{\mathbb{Q}} f^*(K_X + \Delta) + \sum_{E_i \text{ exceptional}} a(E_i, X, \Delta) E_i, \\ K_Y &\sim_{\mathbb{Q}} f^*(K_X + \Delta) + \sum_{E_i \text{ arbitrary}} a(E_i, X, \Delta) E_i. \end{aligned}$$

In the second line, $-f_*^{-1}\Delta$ is included in the sum, since

$$a(f_*^{-1}D_i, X, \Delta) = a(D_i, X, \Delta) = -a_i.$$

When the divisors to be intersected are $\mathbb{Q}$ -Cartier, these relations imply numerical equivalence on proper curves. In particular, if f is proper, we may use relative numerical equivalence $\equiv_f$ or $\equiv_X$ on curves contracted by f ; the base X need not be proper. This interpretation does not make each individual Weil divisor E_i or K_Y $\mathbb{Q}$ -Cartier.

Remark 5.2.10. By choosing compatible representatives of the canonical divisors, we obtain the equality of $\mathbb{Q}$ -Weil divisors

$$K_Y + f_*^{-1}\Delta = f^*(K_X + \Delta) + \sum_{E_i \text{ exceptional}} a(E_i, X, \Delta)E_i.$$

This was used in the proof of generalized Kawamata–Viehweg vanishing. The equality is meaningful because $K_X + \Delta$ is $\mathbb{Q}$ -Cartier; it does not require every term on the left to be $\mathbb{Q}$ -Cartier separately.

Lemma 5.2.11. *With notation as above, assume that Δ' is effective and $\mathbb{Q}$ -Cartier. Then*

$$a(E, X, \Delta) \geq a(E, X, \Delta + \Delta')$$

for every divisor E over X , with strict inequality if and only if $\text{center}_X E \subseteq \text{Supp } \Delta'$.

Proof. Choose a common multiple $m > 0$ such that $m(K_X + \Delta)$ and $m\Delta'$ are Cartier and all relevant rational coefficients have integral multiples. On a model carrying E , the local generator formulas give the global equalities, with compatible canonical divisors,

$$\begin{aligned} mK_Y &= mf^*(K_X + \Delta) + \sum_i ma(E_i, X, \Delta)E_i, \\ mK_Y &= mf^*(K_X + \Delta + \Delta') + \sum_i ma(E_i, X, \Delta + \Delta')E_i, \end{aligned}$$

where the sums include non-exceptional divisors. Subtracting gives

$$a(E, X, \Delta) - a(E, X, \Delta + \Delta') = \text{ord}_E(f^*\Delta') \geq 0.$$

The order is positive precisely when the local equation of $m\Delta'$ vanishes at the center of E , namely when $\text{center}_X E \subseteq \text{Supp } \Delta'$. $\square$

For us the most important values are the infima of $a(E, X, \Delta)$ as E runs through various sets of divisors. We use several versions:

Definition 5.2.12 (Discrepancy of a pair). The discrepancy of (X, Δ) is

$$\text{discrep}(X, \Delta) = \inf_E \{a(E, X, \Delta) : E \text{ is an exceptional divisor over } X\}.$$

Thus E runs through all prime exceptional divisors on all birational models of X . Equivalently, the infimum is over divisorial valuations whose center on X is non-empty and has codimension at least two. For $\Delta = 0$, write $\text{discrep}(X)$.

The total discrepancy is

$$\text{totaldiscrep}(X, \Delta) = \inf_E \{a(E, X, \Delta) : E \text{ is a divisor over } X\}.$$

This also includes the prime divisors on X , or equivalently all divisorial valuations with a non-empty center on X . For $\Delta = 0$, write $\text{totaldiscrep}(X)$.

5.2B Computation of Discrepancies

First we recall a result of Zariski [Zar39, 19.10], which shows that every divisorial valuation with a center can be reached by a sequence of blow-ups. The proof follows [Art86, 5.2].

Lemma 5.2.13. *Let X be an algebraic variety over an arbitrary field k , not necessarily algebraically closed, and in any positive dimension. Let $(R, \mathfrak{m}_R)$ be a discrete valuation ring of $k(X)$, containing k , such that*

$$\text{trdeg}_k(R/\mathfrak{m}_R) = \dim X - 1.$$

Assume that R has a center on X : there is a morphism $f : Y = \text{Spec } R \rightarrow X$ which induces the identity on the function field. Let $y \in Y$ be the closed point. Define a sequence as follows:

- *Set $X_0 = X$ and $f_0 = f$.*
- *If $f_i : Y \rightarrow X_i$ is defined, let $Z_i = \overline{\{x_i\}}$, where $x_i = f_i(y)$. Set $X_{i+1} = \text{Bl}_{Z_i} X_i$, and let $f_{i+1} : Y \rightarrow X_{i+1}$ be the induced morphism.*

Then f_n induces an isomorphism $\mathcal{O}_{X_n, x_n} \cong R$ for some $n \geq 0$.

Proof. The normalized valuation of the discrete valuation ring is

$$v_R(a) = j \quad \text{if } a = t^j u, \quad j \in \mathbb{Z},$$

where t is a uniformizer and u a unit.

Step 1: Suppose that $f_n : Y \rightarrow X_n$ is defined and maps y to x_n . The point x_n need not be closed. Dominance gives an inclusion

$$\mathcal{O}_n := \mathcal{O}_{X_n, x_n} \hookrightarrow R,$$

which is local; write $\mathfrak{m}_n = \mathcal{O}_n \cap \mathfrak{m}_R$. Choose generators $z_1, \dots, z_r$ of $\mathfrak{m}_n$, ordered so that

$$v_R(z_1) \leq v_R(z_2) \leq \dots \leq v_R(z_r).$$

Set

$$\mathcal{O}'_n = \mathcal{O}_n[z_2/z_1, \dots, z_r/z_1].$$

This is the coordinate ring of the chart $D_+(z_1)$ of the blow-up of $\text{Spec } \mathcal{O}_n$ at its closed point. To see its relation to X_{n+1} , take the natural flat morphism $g : \text{Spec } \mathcal{O}_n \rightarrow X_n$. We have the Cartesian diagram

$$\begin{array}{ccc} X_{n+1} \times_{X_n} \text{Spec } \mathcal{O}_n & \longrightarrow & X_{n+1} \\ \downarrow & & \downarrow \\ \text{Spec } \mathcal{O}_n & \xrightarrow{g} & X_n \end{array}$$

By base change for relative Proj [EGA II, 3.5.3; GW20, 13.27],

$$\begin{aligned} X_{n+1} \times_{X_n} \text{Spec } \mathcal{O}_n &\cong \text{Proj}_{\mathcal{O}_n} \left(g^* \bigoplus_{d \geq 0} \mathcal{I}_{Z_n}^d \right) \\ &\cong \text{Proj}_{\mathcal{O}_n} \bigoplus_{d \geq 0} \mathfrak{m}_n^d = \text{Bl}_{\mathfrak{m}_n} \text{Spec } \mathcal{O}_n. \end{aligned}$$

Here flatness identifies $g^* \mathcal{I}_{Z_n}^d$ with $\mathfrak{m}_n^d$. The degree-zero part of the localized graded algebra on $D_+(z_1)$ is precisely

$$\left(\left(\bigoplus_{d \geq 0} \mathfrak{m}_n^d \right)_{z_1} \right)_0 = \mathcal{O}_n[z_2/z_1, \dots, z_r/z_1].$$

Every non-zero ideal of R is principal, so the inverse image of $\mathcal{I}_{Z_n}$ on Y is invertible. The universal property of blowing up gives f_{n+1} . Moreover,

$$v_R(z_i/z_1) = v_R(z_i) - v_R(z_1) \geq 0,$$

so $\mathcal{O}'_n \subseteq R$ and x_{n+1} lies in this chart. Consequently $\mathcal{O}_{n+1}$ is the localization of $\mathcal{O}'_n$ at $\mathcal{O}'_n \cap \mathfrak{m}_R$.

Step 2: We claim that $R = \bigcup_{n \geq 0} \mathcal{O}_n$. We already have $\mathcal{O}_n \subseteq R$. Take $u \in R \setminus \mathcal{O}_n$. Since the blow-ups are birational,

$$k(X) = k(X_n) = \text{Frac } \mathcal{O}_n,$$

so write $u = y_1/y_2$ with $y_i \in \mathcal{O}_n$. The denominator cannot be a unit, since otherwise $u \in \mathcal{O}_n$. Therefore

$$v_R(y_2) > 0, \quad v_R(y_1) \geq v_R(y_2) > 0,$$

the second inequality following from $u \in R$. Thus both y_i lie in $\mathfrak{m}_n$. Write

$$y_i = \sum_{j=1}^r z_j y_{ij}, \quad y_{ij} \in \mathcal{O}_n,$$

and set

$$y'_i = y_i/z_1 = y_{i1} + (z_2/z_1)y_{i2} + \cdots + (z_r/z_1)y_{ir} \in \mathcal{O}'_n \subseteq \mathcal{O}_{n+1}.$$

Then $u = y'_1/y'_2$ and

$$v_R(y'_2) = v_R(y_2) - v_R(z_1) < v_R(y_2).$$

If u is still not in $\mathcal{O}_{n+1}$, both numerator and denominator again belong to its maximal ideal and we repeat. The non-negative integer valuation of the denominator strictly decreases. After finitely many steps it reaches zero, so the denominator is a unit in the corresponding local ring. Hence $u \in \mathcal{O}_{n+s}$ for some s , proving the claim.

Step 3: Choose $t_1, \dots, t_r \in R$ whose residues form a transcendence basis of $R/\mathfrak{m}_R$ over $\bar{k}$. By Step 2, all t_i belong to $\mathcal{O}_n$ for sufficiently large n . Its residue field $k(x_n) = \mathcal{O}_n/\mathfrak{m}_n$ embeds into $R/\mathfrak{m}_R$ and contains these algebraically independent residues. Therefore

$$\mathrm{trdeg}_k k(x_n) = \mathrm{trdeg}_k (R/\mathfrak{m}_R) = \dim X - 1 \quad \text{for } n \gg 0. \quad (5.2)$$

Since $\mathcal{O}_n$ is essentially of finite type over k , the dimension formula gives

$$\dim \mathcal{O}_n = \dim X_n - \dim \overline{\{x_n\}} = \mathrm{trdeg}_k k(X) - \mathrm{trdeg}_k k(x_n) = 1.$$

In particular,

$$\mathrm{trdeg}_{k(x_n)} (R/\mathfrak{m}_R) = \mathrm{trdeg}_k (R/\mathfrak{m}_R) - \mathrm{trdeg}_k k(x_n) = 0.$$

Let R' be the integral closure of $\mathcal{O}_n$ in $k(X)$. Since R is integrally closed,

$$\mathcal{O}_n \subseteq R' \subseteq R \subseteq k(X).$$

The normalization is finite for a ring essentially of finite type over a field, so R' is a finite $\mathcal{O}_n$ -module of dimension one. Set $\mathfrak{p} = R' \cap \mathfrak{m}_R$. Then $R'_\mathfrak{p}$ is a one-dimensional normal local ring, hence a discrete valuation ring, and it is dominated by R . Two discrete valuation rings with the same fraction field cannot properly dominate one another, so $R'_\mathfrak{p} = R$.

Choose finitely many $\mathcal{O}_n$ -module generators $v_1, \dots, v_s$ of R' . Step 2 puts all of them in $\mathcal{O}_k$ for sufficiently large k , so $R' \subseteq \mathcal{O}_k \subseteq R$. Every element of $R' \setminus \mathfrak{p}$ is outside the maximal ideal of $\mathcal{O}_k$, hence invertible there. Thus

$$R = R'_\mathfrak{p} \subseteq \mathcal{O}_k \subseteq R,$$

which proves $\mathcal{O}_k = R$. □

We explain the geometrical meaning of the lemma.

Remark 5.2.14. After finitely many blow-ups, we obtain a point x_n with

$$\dim \mathcal{O}_{X_n, x_n} = \operatorname{codim}_{X_n} \overline{\{x_n\}} = 1.$$

Thus $Z_n = \overline{\{x_n\}}$ is the desired prime divisor. The local ring there equals the prescribed valuation ring.

$$\begin{array}{ccccccc}
 & & & & \operatorname{trdeg}_k k(x_n) \\
 & & & & = \dim X - 1 \\
 X & \longleftarrow & X_1 & \longleftarrow & X_2 & \longleftarrow & \cdots & \longleftarrow & X_n \ni x_n \\
 & \nwarrow f & & & & \nearrow f_n & & & \\
 & & y \in \operatorname{Spec} R & & & & & & \\
 & & \operatorname{trdeg}_k k(y) = \dim X - 1 & & & & & &
 \end{array}$$

We return to characteristic zero. The next two lemmas make it possible to compute discrepancies in many cases. The first shows what happens under one blow-up, and the second shows how to use this step inductively. Lemma 5.2.13 ensures that all divisorial valuations with centers are covered by this method.

Lemma 5.2.15. *Let X be a normal variety, and $\Delta = \sum_i a_i D_i$ a sum of distinct prime divisors such that $K_X + \Delta$ is $\mathbb{Q}$ -Cartier. Let $Z \subseteq X$ be an integral closed subvariety of codimension $k \geq 2$ not contained in $\operatorname{Sing}(X)$. Let $p : \operatorname{Bl}_Z X \rightarrow X$ be its blow-up, and let E be the unique exceptional component dominating Z . This blow-up is smooth near the generic point of E ; if necessary, normalize it when regarding E as a divisor over X . Then*

$$a(E, X, \Delta) = k - 1 - \sum_i a_i \operatorname{mult}_Z D_i.$$

Proof. Step 1: If Z is singular, $\operatorname{Bl}_Z X$ need not be normal even when X is smooth. Take

$$U = X_{\operatorname{reg}} \setminus \operatorname{Sing}(Z), \quad Z_U = Z \cap U.$$

Then Z_U is a smooth closed subvariety of the smooth variety U , and it contains the generic point of Z . There is a Cartesian diagram

$$\begin{array}{ccc}
 \operatorname{Bl}_Z X \times_X U & \longrightarrow & \operatorname{Bl}_Z X \\
 \downarrow & & \downarrow \\
 U & \xrightarrow{j} & X
 \end{array}$$

By base change for Proj,

$$\mathrm{Bl}_Z X \times_X U \cong \mathrm{Proj}_U \bigoplus_{d \geq 0} j^* \mathcal{I}_Z^d \cong \mathrm{Proj}_U \bigoplus_{d \geq 0} \mathcal{I}_{Z_U}^d = \mathrm{Bl}_{Z_U} U.$$

This is smooth. Its exceptional divisor is irreducible, since Z_U is integral. The generic point of E lies here, and

$$\mathcal{O}_{\mathrm{Bl}_Z X, \eta_E} = \mathcal{O}_{\mathrm{Bl}_{Z_U} U, \eta_E}$$

is a discrete valuation ring.

Step 2: We compute on this open set, writing $E_U = E \cap \mathrm{Bl}_{Z_U} U$. The canonical divisor formula for the blow-up of a smooth center gives

$$K_{\mathrm{Bl}_{Z_U} U} - p^* K_U = (k-1)E_U.$$

Also, each $D_i|_U$ is Cartier, and

$$p^*(D_i|_U) = p_*^{-1}(D_i|_U) + \mathrm{mult}_Z D_i E_U.$$

Consequently

$$\begin{aligned} a(E_U, U, \Delta|_U) E_U &= (K_{\mathrm{Bl}_{Z_U} U} - p^* K_U) - (p^*(\Delta|_U) - p_*^{-1}(\Delta|_U)) \\ &= \left(k-1 - \sum_i a_i \mathrm{mult}_Z D_i \right) E_U. \end{aligned}$$

The valuation and discrepancy are unchanged on this neighbourhood of the generic point, so this is also $a(E, X, \Delta)$. $\square$

Lemma 5.2.16. *Let $f : Y \rightarrow X$ be a proper birational morphism of normal varieties. Let Δ_Y and Δ_X be $\mathbb{Q}$ -divisors such that both $K_Y + \Delta_Y$ and $K_X + \Delta_X$ are $\mathbb{Q}$ -Cartier, and*

$$K_Y + \Delta_Y \equiv_f f^*(K_X + \Delta_X), \quad f_* \Delta_Y = \Delta_X.$$

Then, for any divisor F over X ,

$$a(F, Y, \Delta_Y) = a(F, X, \Delta_X).$$

Before giving the proof, we need the Negativity Lemma; see [Bir12, 4.15].

Lemma 5.2.17 (Negativity Lemma). *Let $\psi : Y \rightarrow X$ be a proper birational morphism of normal varieties. Let D be a $\mathbb{Q}$ -Cartier divisor on Y such that*

(1) $-D$ is ψ -nef;

(2) $\psi_* D \geq 0$.

Then $D \geq 0$.

Proof. Step 1: By Chow's lemma, followed by normalization, there is a proper birational morphism $\mu : Y' \rightarrow Y$ such that $\psi' = \psi \circ \mu : Y' \rightarrow X$ is projective. Set $D' = \mu^*D$. For every ψ' -contracted curve C , the projection formula gives

$$(-D' \cdot C)_{Y'} = (-D \cdot \mu_*C)_Y \geq 0.$$

Also,

$$\psi'_*D' = \psi_*(\mu_*\mu^*D) = \psi_*D \geq 0,$$

because $\mu_*\mu^*D = D$: the strict transform pushes forward to D , and the exceptional terms push forward to zero. If $D' \geq 0$, then $D = \mu_*D' \geq 0$. We may therefore assume that ψ is projective.

Step 2: The assertion is local on X , so we may work over an affine open subset of X meeting the image of any proposed negative component. The negative part of D is ψ -exceptional, since $\psi_*D \geq 0$. We will construct an effective divisor whose negative is *relatively ample*, and use a contracted curve on a negative component directly.

It is not enough to cut Y by arbitrary general hyperplanes and assert that the restricted divisor still has effective pushforward to the image surface: an exceptional divisor of $Y \rightarrow X$ may become non-exceptional after such a cut. Working over an affine base avoids this problem.

Step 3: If desired, resolve Y first; pullback preserves the two hypotheses exactly as in Step 1, and effectivity can be pushed back down. We may thus assume that Y is smooth. Choose an effective ψ -very ample Cartier divisor H_Y which contains no exceptional prime divisor. Since X is affine, we can choose an effective principal Cartier divisor H_X whose support contains $\psi(H_Y)$ and the images of all exceptional prime divisors. Taking its defining function to a sufficiently high power ensures

$$E := \psi^*H_X - H_Y \geq 0.$$

Every exceptional prime divisor occurs in E with positive coefficient, and $-E$ is ψ -ample because it is relatively linearly equivalent to H_Y . In particular, for every contracted integral curve C ,

$$(\psi^*H_X \cdot C) = 0, \quad (-E \cdot C) = (H_Y \cdot C) > 0. \quad (5.3)$$

Here E is allowed to have non-exceptional components; only its effectiveness and positivity on every negative component of D are needed.

Step 4: Suppose that D is not effective. Choose the smallest non-negative rational number e for which $D + eE \geq 0$. Explicitly,

$$e = \max_{\text{coeff}_P D < 0} \frac{-\text{coeff}_P D}{\text{coeff}_P E} > 0.$$

There is an exceptional prime divisor P with negative coefficient in D and coefficient zero in $D + eE$. Choose a general closed point of P outside $\text{Supp}(D + eE)$.

Since P is exceptional, its general fibre over $\psi(P)$ has positive dimension; it is projective. By cutting this fibre with hyperplanes through the chosen point, we obtain a contracted integral curve $C \subseteq P$ which is not contained in $\text{Supp}(D + eE)$.

The effective $\mathbb{Q}$ -Cartier divisor $D + eE$ therefore satisfies $(D + eE) \cdot C \geq 0$. On the other hand, $-D$ is ψ -nef and (5.3) gives

$$(D + eE) \cdot C = D \cdot C + e(E \cdot C) < 0,$$

a contradiction. Hence $D \geq 0$. □

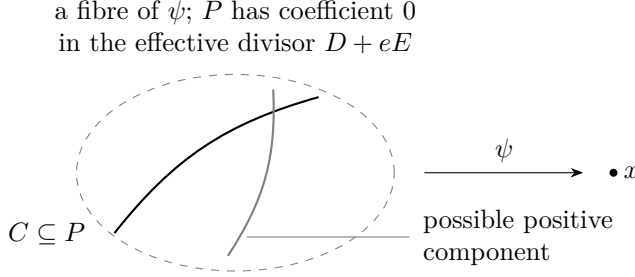


Figure 5.5: A contracted curve C not contained in $\text{Supp}(D + eE)$. Any intersections with its positive components have non-negative degree.

We now prove Lemma 5.2.16.

Proof of Lemma 5.2.16. Our strategy is to use the Negativity Lemma to show that a divisor is zero, and then compare coefficients.

Step 1: Choose compatible canonical divisors, so that $f_*K_Y = K_X$, and set

$$D = (K_Y + \Delta_Y) - f^*(K_X + \Delta_X).$$

This is $\mathbb{Q}$ -Cartier. Pushing forward gives

$$\begin{aligned} f_*D &= f_*(K_Y + \Delta_Y) - f_*f^*(K_X + \Delta_X) \\ &= (K_X + \Delta_X) - (K_X + \Delta_X) = 0. \end{aligned}$$

Moreover $D \equiv_f 0$, so both D and $-D$ are f -nef. Applying the Negativity Lemma to D and $-D$ gives $D \geq 0$ and $-D \geq 0$, whence $D = 0$.

Step 2: Given a divisor F over X , take a common birational model $g : W \rightarrow Y$ on which F appears, and write $h = f \circ g$. Properness of f ensures that the valuation has a center on Y . With compatible canonical divisors,

$$\begin{aligned} K_W &= g^*(K_Y + \Delta_Y) + \sum_i a(E_i, Y, \Delta_Y) E_i, \\ K_W &= h^*(K_X + \Delta_X) + \sum_i a(E_i, X, \Delta_X) E_i, \end{aligned}$$

where the sums range over the prime divisors on W with non-zero coefficients. Since

$$g^*(K_Y + \Delta_Y) - h^*(K_X + \Delta_X) = g^*D = 0,$$

comparison of the coefficient of F proves the assertion. $\square$

Remark 5.2.18. If we assume only the relative numerical equivalence in Lemma 5.2.16, subtraction gives merely

$$\sum_i (a(E_i, Y, \Delta_Y) - a(E_i, X, \Delta_X)) E_i \equiv_h 0$$

on a common model. Numerical triviality by itself does not identify the coefficients. The exceptional support and the Negativity Lemma are needed to deduce the divisor equality. If the stronger equality $K_Y + \Delta_Y = f^*(K_X + \Delta_X)$ is assumed from the outset, then the comparison in Step 2 is immediate and no Negativity Lemma is needed.

Example 5.2.19. For a proper birational morphism $f : Y \rightarrow X$ of normal varieties, choose compatible canonical divisors and write

$$K_Y = f^*(K_X + \Delta_X) + \sum_{E_i \text{ arbitrary}} a(E_i, X, \Delta_X) E_i.$$

Set

$$\begin{aligned} \Delta_Y &= \sum_{E_i \text{ arbitrary}} -a(E_i, X, \Delta_X) E_i \\ &= \sum_{D_i \subseteq \text{Supp } \Delta_X} a_i f_*^{-1} D_i - \sum_{E_j \text{ exceptional}} a(E_j, X, \Delta_X) E_j. \end{aligned}$$

Then $K_Y + \Delta_Y = f^*(K_X + \Delta_X)$ is $\mathbb{Q}$ -Cartier, and $f_* \Delta_Y = \Delta_X$. This is the direct, crepant-pullback case of Lemma 5.2.16.

The following is the first computation of discrepancies of pairs.

Corollary 5.2.20. *Let X be a normal variety of dimension at least two, and $\Delta = \sum_i a_i D_i$ a $\mathbb{Q}$ -divisor such that $K_X + \Delta$ is $\mathbb{Q}$ -Cartier. Then:*

(1) *Either $\text{discrep}(X, \Delta) = -\infty$, or*

$$-1 \leq \text{totaldiscrep}(X, \Delta) \leq \text{discrep}(X, \Delta) \leq 1.$$

(2) *If X is smooth, then $\text{discrep}(X, 0) = 1$.*

(3) *If X is smooth, $\sum_i D_i$ is an snc divisor, and $a_i \leq 1$ for every i , then*

$$\text{discrep}(X, \Delta) = \min \left\{ \min_{i \neq j, D_i \cap D_j \neq \emptyset} (1 - a_i - a_j), \min_i (1 - a_i), 1 \right\}.$$

An infimum over an empty index set is omitted from this minimum.

Proof. **For (1)**

Choose an integral codimension-two subvariety Z whose generic point lies in the smooth locus of X and outside $\text{Supp } \Delta$. We only need Z smooth on a neighbourhood of that generic point. The exceptional divisor E_0 dominating Z on the blow-up has

$$a(E_0, X, \Delta) = 2 - 1 = 1$$

by Lemma 5.2.15. Thus $\text{discrep}(X, \Delta) \leq 1$. The inclusion of the set of exceptional divisors into the set of all divisors over X gives

$$\text{totaldiscrep}(X, \Delta) \leq \text{discrep}(X, \Delta) \leq 1.$$

Suppose now that $\text{totaldiscrep}(X, \Delta) < -1$. There is a divisor E over X with $a(E, X, \Delta) = -1 - c$, where $c > 0$. Take a normal birational model $f : Y \rightarrow X$ carrying E and define

$$K_Y + \Delta_Y = f^*(K_X + \Delta)$$

as in Example 5.2.19. The coefficient of E in Δ_Y is $1 + c$.

Choose a codimension-two subvariety $Z_0 \subseteq E$ whose generic point lies where Y and E are smooth and no other component of Δ_Y is present. Near that point, E is a smooth divisor and $\text{mult}_{Z_0} E = 1$. We may shrink to make this a blow-up of smooth varieties along a smooth center. For $g_1 : Y_1 \rightarrow Y$ with exceptional divisor E_1 , Lemma 5.2.15 gives

$$a(E_1, Y, \Delta_Y) = 1 - (1 + c) = -c.$$

By crepant pullback, $a(E_1, X, \Delta) = -c$ as well.

Let Z_1 be the intersection of E_1 and the strict transform of E . Put

$$\Delta_{Y_1} = g_{1*}^{-1} \Delta_Y + cE_1, \quad K_{Y_1} + \Delta_{Y_1} = g_1^*(K_Y + \Delta_Y).$$

Blowing up Z_1 gives an exceptional divisor E_2 with

$$a(E_2, Y_1, \Delta_{Y_1}) = 1 - c - (1 + c) = -2c.$$

Again this equals $a(E_2, X, \Delta)$. Repeating along the intersection of the strict transform of E with the newest exceptional divisor gives

$$a(E_j, X, \Delta) = -jc \rightarrow -\infty.$$

Therefore $\text{discrep}(X, \Delta) = -\infty$ whenever $\text{totaldiscrep}(X, \Delta) < -1$.

For (2)

This is the special case $\Delta = 0$ of (3); the argument below also shows directly that every exceptional discrepancy over a smooth variety is at least one.

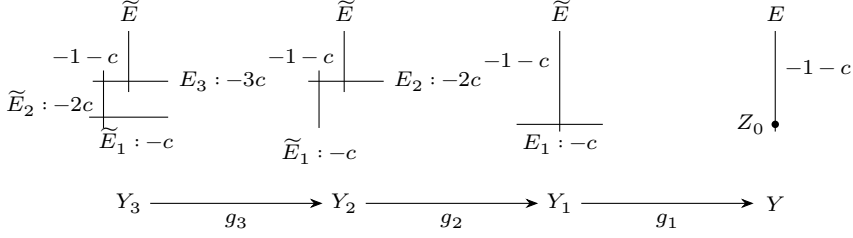


Figure 5.6: Discrepancies of the exceptional divisors under successive blow-ups.

For (3)

Set

$$r(X, \Delta) = \min \left\{ \min_{i \neq j, D_i \cap D_j \neq \emptyset} (1 - a_i - a_j), \quad \min_i (1 - a_i), \quad 1 \right\}.$$

First we construct blow-ups realizing the terms in this minimum:

- (1) Blow up an irreducible component Z of $D_i \cap D_j$. At its generic point no other boundary component is present, and

$$a(E, X, \Delta) = 2 - 1 - a_i - a_j = 1 - a_i - a_j.$$

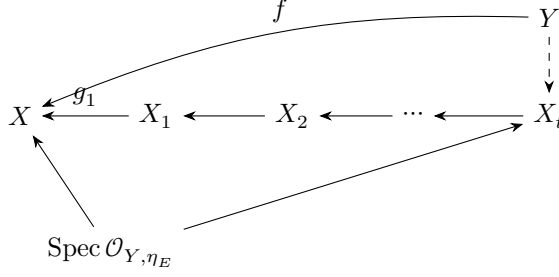
- (2) Blow up a general codimension-two subvariety $Z \subseteq D_i$, not contained in another D_j . Then

$$a(E, X, \Delta) = 2 - 1 - a_i = 1 - a_i.$$

- (3) Blow up a codimension-two center generically outside $\text{Supp } \Delta$. Then $a(E, X, \Delta) = 1$.

These centers can be taken smooth near their generic points; discrepancy is determined there. Hence $\text{discrep}(X, \Delta) \leq r(X, \Delta)$.

For the converse, let E be an exceptional divisor over X . By Lemma 5.2.13, its valuation is obtained after a finite succession of blow-ups of its centers. Near the generic point of each center, we may shrink so that the center is smooth and has normal crossings with the current snc boundary. Indeed, the boundary components containing the center define a smooth stratum, and the center is generically smooth inside that stratum; the remaining components can be avoided. Shrinking cannot decrease the displayed minimum r , since it only removes terms.



Induct on the number t of blow-ups. Let the first center have codimension $k \geq 2$ and be contained in exactly $D_1, \dots, D_b$. Since the support is snc, $D_1 \cap \dots \cap D_b$ has codimension b and contains the center, so $b \leq k$. For the first exceptional divisor E_1 ,

$$e := a(E_1, X, \Delta) = k - 1 - \sum_{l=1}^b a_l.$$

There are three cases:

- (1) If $b = 0$, then $e = k - 1 \geq 1 \geq r(X, \Delta)$.
- (2) If $b = 1$, then $e = k - 1 - a_1 \geq 1 - a_1 \geq r(X, \Delta)$.
- (3) If $b \geq 2$, then

$$\begin{aligned} e &= (k - b - 1) + \sum_{l=1}^b (1 - a_l) \\ &\geq -1 + (1 - a_1) + (1 - a_2) = 1 - a_1 - a_2 \geq r(X, \Delta), \end{aligned}$$

since every $a_l \leq 1$.

Thus the case $t = 1$ is proved.

Define

$$\Delta_1 = g_{1*}^{-1} \Delta - eE_1, \quad K_{X_1} + \Delta_1 = g_1^*(K_X + \Delta).$$

The support is snc on the chosen neighbourhood. Since $e \geq r(X, \Delta) \geq -1$, the new coefficient $-e$ is at most one. In computing $r(X_1, \Delta_1)$, the terms involving only strict transforms already occurred in $r(X, \Delta)$. The new terms are $1 + e$ and, for components meeting E_1 , $1 + e - a_i$. Each is at least e , because $a_i \leq 1$. Therefore

$$r(X_1, \Delta_1) \geq \min\{r(X, \Delta), e\} = r(X, \Delta).$$

If E is not E_1 , the induction hypothesis on the remaining blow-ups and Lemma 5.2.16 give

$$a(E, X, \Delta) = a(E, X_1, \Delta_1) \geq r(X_1, \Delta_1) \geq r(X, \Delta).$$

This completes the proof. □

Under suitable conditions, discrepancy can be computed from the exceptional divisors on a fixed resolution:

Corollary 5.2.21. *Let (X, Δ) be a pair with $K_X + \Delta$ $\mathbb{Q}$ -Cartier, and let $f : Y \rightarrow X$ be a resolution, with exceptional prime divisors E_i .*

(1) *In the case $\Delta = 0$, assume K_X is $\mathbb{Q}$ -Cartier and*

$$0 \leq \min_i a(E_i, X) \leq 1.$$

Then $\text{discrep}(X) = \min_i a(E_i, X)$.

(2) *Write $\Delta = \sum_j a_j D_j$, where $a_j \leq 1$. There is a log resolution for which $\sum_j f_*^{-1} D_j$ is smooth, i.e. its components are smooth and pairwise disjoint. For any such resolution, if $a(E_i, X, \Delta) \geq -1$ for every i , then*

$$\text{discrep}(X, \Delta) = \min \left\{ \min_i a(E_i, X, \Delta), \min_j (1 - a_j), 1 \right\}.$$

Empty minima are omitted; the assumption in (1) in particular requires at least one exceptional divisor.

Proof. For (1)

Define Δ_Y by $K_Y + \Delta_Y = f^* K_X$. Then

$$\Delta_Y = - \sum_i a(E_i, X) E_i \leq 0.$$

Monotonicity and Corollary 5.2.20(2) imply

$$\text{discrep}(Y, \Delta_Y) \geq \text{discrep}(Y, 0) = 1.$$

Every exceptional valuation over X is either one of the E_i or exceptional over Y , so

$$\begin{aligned} \text{discrep}(X) &= \min \{ \text{discrep}(Y, \Delta_Y), \min_i a(E_i, X) \} \\ &= \min_i a(E_i, X). \end{aligned}$$

For (2)

Start with a log resolution $g : Z \rightarrow X$, so that the union of the strict transforms of the D_j and the exceptional divisors is snc. Whenever two strict transforms meet, blow up their whole intersection (a disjoint union of smooth codimension-two subvarieties). Their new strict transforms are disjoint, the total boundary remains snc, and pairs which were disjoint stay disjoint. Thus the number of intersecting pairs of strict transforms decreases. Repeating finitely many

times gives a resolution $f : Y \rightarrow X$ with pairwise disjoint strict transforms of the D_j .

Set $b_i = -a(E_i, X, \Delta) \leq 1$ and

$$\Delta_Y = \sum_j a_j f_*^{-1} D_j + \sum_i b_i E_i, \quad K_Y + \Delta_Y = f^*(K_X + \Delta).$$

By Corollary 5.2.20, $\text{discrep}(Y, \Delta_Y)$ is the minimum of the applicable numbers

$$1 - b_i - a_j, \quad 1 - b_i - b_{i'}, \quad 1 - a_j, \quad 1 - b_i, \quad 1,$$

where the first two types require the corresponding divisors to meet. There is no term $1 - a_j - a_{j'}$, since the strict transforms are disjoint. Moreover,

$$\begin{aligned} \text{discrep}(X, \Delta) &= \min\{\text{discrep}(Y, \Delta_Y), \min_i a(E_i, X, \Delta)\} \\ &= \min\{\text{discrep}(Y, \Delta_Y), \min_i (-b_i)\}. \end{aligned}$$

Since all $b_i, a_j \leq 1$,

$$-b_i \leq 1 - b_i - b_{i'}, \quad -b_i \leq 1 - b_i - a_j, \quad -b_i \leq 1 - b_i.$$

All terms involving an exceptional coefficient can therefore be removed from the minimum, leaving

$$\text{discrep}(X, \Delta) = \min\{\min_i (-b_i), \min_j (1 - a_j), 1\}.$$

□

5.2C Singularities for the Pair

Next we define the six classes of singularities which are most important for the Minimal Model Program.

Definition 5.2.22 (Singularities for pairs). Let X be a normal variety and $\Delta = \sum_i a_i D_i$ a sum of distinct prime divisors with $a_i \in \mathbb{Q}$. Assume that $K_X + \Delta$ is $\mathbb{Q}$ -Cartier. We say that (X, Δ) is

$$\begin{array}{ll} \text{terminal} & \text{if } \text{discrep}(X, \Delta) > 0, \\ \text{canonical} & \text{if } \text{discrep}(X, \Delta) \geq 0, \\ \text{klt} & \text{if } \text{discrep}(X, \Delta) > -1 \text{ and } \lfloor \Delta \rfloor \leq 0, \\ \text{plt} & \text{if } \text{discrep}(X, \Delta) > -1, \\ \text{lc} & \text{if } \text{discrep}(X, \Delta) \geq -1. \end{array}$$

Here klt stands for “Kawamata log terminal”, plt for “purely log terminal”, and lc for “log canonical”.

If (X, Δ) is lc, then $a_i \leq 1$ for every i . Otherwise, at a general smooth point of a component with $a_i > 1$, the blow-up construction in Corollary 5.2.20(1) gives exceptional divisors E_j such that

$$a(E_j, X, \Delta) = j(1 - a_i).$$

Here one first blows up a codimension-two center in D_i , then successively the intersection of its strict transform with the newest exceptional divisor:

$$\begin{array}{ccccccc} X_3 & \xrightarrow{g_3} & X_2 & \xrightarrow{g_2} & X_1 & \xrightarrow{g_1} & X \\ E_3 & & E_2 & & E_1 & & D_i \\ 3(1 - a_i) & & 2(1 - a_i) & & 1 - a_i & & -a_i \end{array}$$

Since $1 - a_i < 0$, eventually $j(1 - a_i) < -1$, contradicting log canonicity.

Definition 5.2.23 (Divisorial log terminal). Let X be a normal variety and $\Delta = \sum_i a_i D_i$ a sum of distinct prime divisors, with rational coefficients $0 \leq a_i \leq 1$. Assume that $K_X + \Delta$ is $\mathbb{Q}$ -Cartier. We say that (X, Δ) is dlt, or divisorial log terminal, if and only if there is a closed subset $Z \subseteq X$ such that

- (1) $X \setminus Z$ is smooth and $\Delta|_{X \setminus Z}$ has snc support;
- (2) if E is a divisor over X with center $E \subseteq Z$, then $a(E, X, \Delta) > -1$.

Each of these notions has an important place in the theory of minimal models:

1. Terminal: The Natural Foundation of the MMP

Core Definition: We require $\text{discrep}(X, \Delta) > 0$. For $\Delta = 0$, terminal singularities give the standard class in which to run the MMP starting from smooth varieties.

Geometric Motivation: Suppose we start with a smooth projective variety X . A divisorial contraction or a flip may produce a singular variety X' . Every exceptional divisor over a smooth variety has positive discrepancy. The preservation theorems for the MMP ensure that a $\mathbb{Q}$ -factorial terminal starting variety again gives $\mathbb{Q}$ -factorial terminal varieties after these operations. Thus terminal singularities allow us to continue the program after smoothness is lost. This preservation property is the point needed here; it is not a proof of a set-theoretic minimality assertion about all possible classes of singularities.

2. Canonical: General Type and Canonical Models

Core Definition: We require $\text{discrep}(X, \Delta) \geq 0$. In particular, canonical singularities occur on the canonical models of smooth varieties of general type.

Geometric Motivation: For a smooth variety of general type, we consider the canonical ring

$$R(X) = \bigoplus_{m \geq 0} H^0(X, \mathcal{O}_X(mK_X)).$$

If this ring is finitely generated, we can define $X_{\text{can}} = \text{Proj } R(X)$. For a normal variety X with canonical singularities, pluricanonical forms acquire no poles along exceptional divisors when pulled back to a resolution $f : Y \rightarrow X$. For positive multiples of the index, this gives

$$H^0(X, \mathcal{O}_X(mK_X)) \cong H^0(Y, \mathcal{O}_Y(mK_Y)).$$

This invariance is essential for canonical models. Moduli compactifications require a larger class: for example, KSBA stable varieties may have semi-log-canonical singularities and need not even be normal. They are not restricted to canonical singularities.

3. klt (Kawamata Log Terminal): The Domain of Vanishing Theorems

Core Definition: We require $\text{discrep}(X, \Delta) > -1$ and $\lfloor \Delta \rfloor \leq 0$.

Geometric Motivation: Kawamata–Viehweg vanishing and the Basepoint-Free Theorem naturally apply to klt pairs, together with their respective positivity hypotheses. On a log resolution, the relevant coefficients lie strictly below one, which is the boundary condition underlying the vanishing argument. At discrepancy -1 , the standard klt form of these theorems need not apply; this does not mean that every vanishing theorem fails for lc pairs.

Why introduce a larger class for induction? Adjunction to a divisor takes the form

$$(K_X + S + B)|_S = K_S + B_S.$$

If the pair $(X, S + B)$ is required to be klt, the coefficient of S cannot be one. Thus klt pairs do not directly include the coefficient-one boundary needed in this adjunction setup. This motivates plt pairs; it does not rule out inductive arguments involving klt varieties by other methods or by perturbing the boundary.

4. plt (Purely Log Terminal): Adjunction and Induction

Core Definition: We allow coefficients equal to one in the boundary, but require $a(E, X, \Delta) > -1$ for every exceptional divisor E .

Geometric Motivation: If $(X, S + B)$ is plt, with $B \geq 0$ and S a prime divisor of coefficient one, then S is normal. Adjunction produces a boundary B_S , the different, such that

$$(K_X + S + B)|_S = K_S + B_S,$$

and (S, B_S) is klt. In the smooth transverse case $B_S = B|_S$, but in general it includes contributions from the singularities. Together with suitable positivity hypotheses, vanishing and restriction sequences can then relate sections on X and S . The basic sequence is

$$0 \rightarrow \mathcal{O}_X(-S) \rightarrow \mathcal{O}_X \rightarrow \mathcal{O}_S \rightarrow 0;$$

when S is not Cartier, $\mathcal{O}_X(-S)$ here is its divisorial ideal sheaf. Thus the plt condition supplies a useful setup for induction on dimension.

5. lc (Log Canonical) and dlt: The Log Canonical Boundary

Core Definition: Log canonicity requires $\text{discrep}(X, \Delta) \geq -1$. Discrepancies are still defined below this range, but the infimum then equals $-\infty$.

Geometric Motivation: As proved above, once a divisor with discrepancy less than -1 appears, further blow-ups yield discrepancies tending to $-\infty$. The lc condition excludes this behaviour. It nevertheless permits complicated singularities, and the standard klt vanishing and contraction arguments cannot simply be applied without further work.

The dlt class retains lc behaviour while requiring discrepancies over the bad locus Z in Definition 5.2.23 to be strictly greater than -1 . Exceptional divisors of discrepancy -1 are still allowed over the snc boundary strata of $X \setminus Z$. For example, blowing up the intersection of two coefficient-one boundary components on a smooth surface gives discrepancy -1 , although the original snc pair is dlt. Thus dlt does not prohibit all exceptional divisors of discrepancy -1 . It gives a controlled setting for boundary strata and adjunction.

Remark 5.2.24 (Discrepancy in the $\mathbb{R}$ -Cartier case). For a pair with $K_X + \Delta$ $\mathbb{Q}$ -Cartier, we defined discrepancy using the vanishing order of a pulled-back local generator after clearing denominators. This line-bundle description applies directly to rational coefficients. In algebraic geometry, we can also treat $K_X + \Delta$ $\mathbb{R}$ -Cartier by using divisor equalities.

Choose compatible canonical divisors and write

$$K_Y + f_*^{-1}\Delta = f^*(K_X + \Delta) + \sum_{E_i \text{ exceptional}} a(E_i, X, \Delta)E_i.$$

Equivalently, including all prime divisors on Y in the sum,

$$\sum_i a(E_i, X, \Delta)E_i = K_Y - f^*(K_X + \Delta).$$

This defines the real discrepancy $a(E, X, \Delta)$, independently of the model. We also set

$$\begin{aligned} \text{discrep}(X, \Delta) &= \inf\{a(E, X, \Delta) : E \text{ exceptional over } X\}, \\ \text{totaldiscrep}(X, \Delta) &= \inf\{a(E, X, \Delta) : E \text{ a divisor over } X\}. \end{aligned}$$

The previous arguments which use divisor equalities, orders and inequalities remain valid for real coefficients. This includes Lemmas 5.2.11 and 5.2.15–5.2.17 and Corollaries 5.2.20–5.2.21. In the Negativity Lemma, the same proof uses the minimum real number e , rather than a rational one. The definitions of terminal, canonical, klt, plt, lc and dlt extend accordingly.

The following is a Bertini-type comparison between special and general members of a linear series.

Corollary 5.2.25. *Let X be a proper normal variety, $\Delta \geq 0$ an effective $\mathbb{R}$ -divisor with $K_X + \Delta$ $\mathbb{R}$ -Cartier, and $|L|$ a non-empty linear series of effective Cartier divisors. Let $L_0 \in |L|$ and let $L_g \in |L|$ be a general member. Then, for every $c \geq 0$,*

$$\text{discrep}(X, \Delta + cL_0) \leq \text{discrep}(X, \Delta + cL_g).$$

Reduction to the rational Bertini theorem. For rational coefficients, we use the Bertini discrepancy comparison of [KM98, Corollary 2.33], with properness and effectivity explicitly imposed here. The proof below explains the passage to real coefficients and the discrepancy bookkeeping on a resolution. The rational Bertini comparison is an external input, rather than a consequence of birational invariance of discrep .

Step 1: Choose a resolution $f : Y \rightarrow X$ such that

- (1) $f^*|L| = B + |F|$, with $B \geq 0$ and $|F|$ free;
- (2) the strict transforms of the components of $\text{Supp } \Delta$ are smooth and pairwise disjoint;
- (3) $\text{Supp } B \cup f_*^{-1} \text{Supp } \Delta \cup \text{Ex}(f)$ is snc.

For $L_\lambda \in |L|$, write $f^*L_\lambda = B + F_\lambda$ and

$$K_Y + \Delta_Y = f^*(K_X + \Delta), \quad \Gamma = \Delta_Y + cB.$$

Then

$$f^*(K_X + \Delta + cL_\lambda) = K_Y + \Gamma + cF_\lambda.$$

For general F_g , the support upstairs is snc. The conditions on a general member depend on the fixed supports, not on their coefficients.

Step 2: Let E_i be the f -exceptional prime divisors, and set

$$C(\lambda) = \max_i \text{coeff}_{E_i}(\Gamma + cF_\lambda).$$

If there is no exceptional divisor, omit the term $-C(\lambda)$ below. By distinguishing divisors already on Y from those exceptional over Y , the correct identity is

$$\text{discrep}(X, \Delta + cL_\lambda) = \min\{\text{discrep}(Y, \Gamma + cF_\lambda), -C(\lambda)\}. \quad (5.4)$$

Individual discrepancies are preserved by the crepant equality in Step 1, but the sets defining the two infima differ. Thus the comparison to prove on this model is

$$\begin{aligned} & \min\{\text{discrep}(Y, \Gamma + cF_0), -C(0)\} \\ & \leq \min\{\text{discrep}(Y, \Gamma + cF_g), -C(g)\}. \end{aligned} \quad (5.5)$$

One cannot in general discard the terms $-C(0)$ and $-C(g)$. If $|F| = \{0\}$, all members are the same divisor and the conclusion is immediate.

Step 3: Suppose now that the coefficients are real. If $(X, \Delta + cL_0)$ is not lc, its discrepancy is $-\infty$ by Corollary 5.2.20, and there is nothing to prove. We may assume this special pair is lc.

Fix the finite support of Δ . Choose local Cartier presentations of $K_X + \Delta$ on a finite open cover, and a common log resolution of this support, the base ideal of $|L|$, and L_0 . Choose L_g in the Bertini open set for this data; it can be chosen independently of the coefficients. Refine the resolution for the two fixed members L_0, L_g , separating all strict transforms as in Corollary 5.2.21.

On this fixed resolution, the coefficients of the crepant boundary and the exceptional discrepancies are affine functions of the coefficients of Δ and of c . The local Cartier presentations give rational linear constraints on these parameters. Effectivity, $c \geq 0$, and log canonicity of the special pair give finitely many rational linear inequalities: the strict-transform coefficients must be at most one, and the exceptional discrepancies at least -1 . Hence the given real parameters lie in a rational polyhedron, and can be approximated inside it by rational parameters (Δ_j, c_j) . These satisfy

$$\Delta_j \geq 0, \quad c_j \geq 0, \quad K_X + \Delta_j \text{ is } \mathbb{Q}\text{-Cartier},$$

and $(X, \Delta_j + c_j L_0)$ is lc.

Step 4: Apply the rational Bertini comparison to each approximation:

$$\text{discrep}(X, \Delta_j + c_j L_0) \leq \text{discrep}(X, \Delta_j + c_j L_g).$$

The left side is at least -1 , so every general pair on the right is lc as well. Their coefficient inequalities are closed and pass to the limit, proving that $(X, \Delta + cL_g)$ is lc. On the lc region, Corollary 5.2.21 computes both discrepancies as minima of finitely many affine functions on the chosen resolution. They are therefore continuous there. Passing to the limit proves the required inequality for real coefficients. $\square$

Example 5.2.26 (The assumption of properness is crucial). Take $X = \mathbb{A}^2$ and $L = \mathbb{C}\langle 1, x \rangle \subseteq H^0(X, \mathcal{O}_X) = \mathbb{C}[x, y]$. The parameter space is $\mathbb{P}(L) \cong \mathbb{P}^1$, with $[a : b]$ corresponding to the section $a + bx$. The special member L_0 , with parameter $[1 : 0]$, is the zero divisor. A general member L_g is the line $a + bx = 0$, with $b \neq 0$. Put $\Delta = 0$ and $0 < c \leq 1$. Then

$$\text{discrep}(X, cL_0) = \text{discrep}(X, 0) = 1.$$

For a point $p \in L_g$, its blow-up has exceptional discrepancy

$$a(E, X, cL_g) = 2 - 1 - c \text{mult}_p L_g = 1 - c.$$

In fact $\text{discrep}(X, cL_g) = 1 - c$ by the snc formula. Thus the special discrepancy is strictly greater than the general one. Here a non-zero section on a non-proper variety can be nowhere vanishing, even though general sections of the same system have zeros.

Example 5.2.27 (The assumption of effectivity is crucial). Many references reserve “pair” for an effective boundary, often with coefficients at most one. In our convention, negative coefficients are also allowed, sometimes under the name “sub-pair”. The preceding comparison need not hold for such boundaries.

Take $X = \mathbb{P}^2$, $\Delta = -dH$ with $d > 0$ and H a line, and $|L| = |\mathcal{O}_{\mathbb{P}^2}(1)|$. Let $L_0 = H$, and let L_g be another line meeting H transversely. For $0 \leq c \leq 1$, Corollary 5.2.20 gives

$$\begin{aligned} \text{discrep}(X, \Delta + cL_g) &= \min\{1 + d - c, 1 - c, 1 + d, 1\} = 1 - c, \\ \text{discrep}(X, \Delta + cL_0) &= \min\{1 - c + d, 1\}. \end{aligned}$$

In particular, for $c = 1$,

$$\text{discrep}(X, \Delta + L_0) = \min\{d, 1\} > 0 = \text{discrep}(X, \Delta + L_g).$$

This is the reverse of the inequality in Corollary 5.2.25.

Definition 5.2.28 (Alternative definition for canonical pairs). Let X be a normal variety and Δ an effective $\mathbb{R}$ -divisor. We say that (X, Δ) has canonical singularities if

- (1) $K_X + \Delta$ is $\mathbb{R}$ -Cartier;
- (2) for every resolution $f : Y \rightarrow X$, writing

$$K_Y + C = f^*(K_X + \Delta),$$

the divisor $-C + f_*^{-1}\Delta$ is effective.

Indeed,

$$\begin{aligned} K_Y + f_*^{-1}\Delta &= f^*(K_X + \Delta) + \sum_{E_i \text{ exceptional}} a(E_i, X, \Delta)E_i \\ &= K_Y + C + \sum_i a(E_i, X, \Delta)E_i. \end{aligned}$$

Thus $-C + f_*^{-1}\Delta = \sum_i a(E_i, X, \Delta)E_i$, and the condition on every resolution is exactly $\text{discrep}(X, \Delta) \geq 0$.

Proposition 5.2.29. *When $\Delta = 0$ and K_X is $\mathbb{Q}$ -Cartier, this definition agrees with Definition 5.1.9.*

Proof. Take $m > 0$ divisible by the Cartier index of K_X . We show that effectivity of $-C$ on every resolution is equivalent to the natural equality

$$f_*\mathcal{O}_Y(mK_Y) = \mathcal{O}_X(mK_X).$$

Write

$$mK_Y = f^*(mK_X) + \sum_i ma_i E_i = f^*(mK_X) - mC,$$

where E_i are exceptional. If $-C \geq 0$, the projection formula gives

$$f_* \mathcal{O}_Y(mK_Y) \cong \mathcal{O}_X(mK_X) \otimes f_* \mathcal{O}_Y(-mC).$$

We have $\mathcal{O}_X = f_* \mathcal{O}_Y \subseteq f_* \mathcal{O}_Y(-mC)$. Conversely, a section of the latter is a rational function regular outside the image of the exceptional divisors, a set of codimension at least two. Normality and algebraic Hartogs show that it is regular on X . Thus $f_* \mathcal{O}_Y(-mC) = \mathcal{O}_X$.

Conversely, under this natural equality, the function 1 on every open $U \subseteq X$ is a section of $\mathcal{O}_Y(-mC)$ on $f^{-1}(U)$. The divisor condition for this section gives $\text{div}(1) - mC = -mC \geq 0$. Hence $-C$ is effective. $\square$

Definition 5.2.30 (Alternative definition for terminal pairs). Let X be a normal variety and Δ an effective $\mathbb{R}$ -divisor. We say that (X, Δ) has terminal singularities if

- (1) $K_X + \Delta$ is $\mathbb{R}$ -Cartier;
- (2) for every resolution $f : Y \rightarrow X$, writing $K_Y + C = f^*(K_X + \Delta)$, the divisor $-C + f_*^{-1}\Delta$ is effective;
- (3) its support is the divisorial part of $\text{Ex}(f)$.

As above,

$$-C + f_*^{-1}\Delta = \sum_{E_i \text{ exceptional}} a(E_i, X, \Delta) E_i.$$

Conditions (2)–(3) say that every exceptional discrepancy is strictly positive. To pass from these individual inequalities to $\text{discrep}(X, \Delta) > 0$, use Corollary 5.2.21 on a fixed log resolution with disjoint strict transforms. Blowing up a general codimension-two center in a boundary component first shows that its coefficient is strictly less than one. The finite minimum in that corollary is therefore positive. This also justifies the equivalence for real coefficients.

Proposition 5.2.31. *When $\Delta = 0$ and K_X is $\mathbb{Q}$ -Cartier, this definition agrees with Definition 5.1.10.*

Proof. Choose $m > 0$ divisible by the Cartier index. On a resolution write

$$mK_Y = f^*(mK_X) + \sum_i ma_i E_i = f^*(mK_X) - mC, \quad E = \sum_i E_i.$$

If every $a_i > 0$, then ma_i is a positive integer and hence $ma_i \geq 1$. The projection formula gives

$$f_* \mathcal{O}_Y(mK_Y - E) \cong \mathcal{O}_X(mK_X) \otimes f_* \mathcal{O}_Y \left(\sum_i (ma_i - 1) E_i \right).$$

The sum is effective and exceptional, so the same Hartogs argument as above shows that the last pushforward is $\mathcal{O}_X$.

Conversely, the natural equality of these sheaves implies that the function 1 is a section of $\mathcal{O}_Y(\sum_i (ma_i - 1)E_i)$ over inverse images of open sets. Hence every $ma_i - 1 \geq 0$, so

$$a_i \geq \frac{1}{m} > 0.$$

Thus $-C$ is effective with support equal to the divisorial exceptional locus, as required. $\square$

Definition 5.2.32 (Alternative definition for klt pairs). Let X be a normal variety and Δ an effective $\mathbb{R}$ -divisor. We say that (X, Δ) is klt if

- (1) $K_X + \Delta$ is $\mathbb{R}$ -Cartier;
- (2) every coefficient of Δ is less than one;
- (3) there exists a log resolution $f : Y \rightarrow X$ such that, writing

$$K_Y + C = f^*(K_X + \Delta), \quad C = \sum_j c_j C_j,$$

every coefficient $c_j < 1$.

The equivalence with Definition 5.2.22 for $\Delta \geq 0$ follows from the snc computation and Corollary 5.2.21. When negative coefficients are allowed, the corresponding terminology is sometimes “sub-klt”.

Definition 5.2.33 (Alternative definition for plt pairs). Let X be a normal variety and Δ an effective $\mathbb{R}$ -divisor. We say that (X, Δ) is plt if

- (1) $K_X + \Delta$ is $\mathbb{R}$ -Cartier;
- (2) every coefficient of Δ is at most one;
- (3) for every log resolution $f : Y \rightarrow X$, writing

$$K_Y + C = f^*(K_X + \Delta), \quad C = \sum_j c_j C_j,$$

we have $c_j < 1$ for every f -exceptional component C_j .

This agrees with Definition 5.2.22 by the finite-resolution formula and $-c_j = a(C_j, X, \Delta)$. The quantifier “for every” is essential.

Definition 5.2.34 (Alternative definition for lc pairs). Let X be a normal variety and Δ an effective $\mathbb{R}$ -divisor. We say that (X, Δ) is lc if

- (1) $K_X + \Delta$ is $\mathbb{R}$ -Cartier;
- (2) every coefficient of Δ is at most one;
- (3) there exists a log resolution $f : Y \rightarrow X$ such that, writing

$$K_Y + C = f^*(K_X + \Delta), \quad C = \sum_j c_j C_j,$$

every coefficient $c_j \leq 1$.

Again the snc computation gives the equivalence for $\Delta \geq 0$. With arbitrary negative coefficients, one sometimes uses the term “sub-lc”.

Definition 5.2.35 (Alternative definition for dlt pairs). Let X be a normal variety and Δ an effective $\mathbb{R}$ -divisor. We say that (X, Δ) is dlt if

- (1) $K_X + \Delta$ is $\mathbb{R}$ -Cartier;
- (2) every coefficient of Δ is at most one;
- (3) there exists a log resolution $f : Y \rightarrow X$ such that, writing

$$K_Y + C = f^*(K_X + \Delta), \quad C = \sum_j c_j C_j,$$

we have $c_j < 1$ for every f -exceptional component C_j .

The equivalence with Definition 5.2.23 is due to Szabó; see [Sza94]. In contrast to the plt definition above, only the existence of one such log resolution is required.

Example 5.2.36. Take $X = \mathbb{A}^n$, $n \geq 2$, with coordinate hyperplanes $D_1, \dots, D_n$, and let $\Delta = \sum_i a_i D_i$ be effective. Then (X, Δ) is

- klt if and only if $0 \leq a_i < 1$ for all i ;
- plt if and only if $0 \leq a_i \leq 1$ for all i , with $a_i = 1$ for at most one i ;
- lc if and only if $0 \leq a_i \leq 1$ for all i .

Here lc and dlt coincide: the identity is already a log resolution with no exceptional divisors.

Example 5.2.37. Take $X = \mathbb{A}^2$ with coordinates x, y and prime divisor $D = \text{div}(x^2 - y^3)$. We determine when (X, dD) is klt or lc, for $0 \leq d \leq 1$.

Construct a log resolution as in [Har77, V, 3.9.1]. First blow up $p_0 = (0, 0)$:

$$f_1 : Y_1 \longrightarrow X.$$

The exceptional divisor is $E_1 \cong \mathbb{P}^1$. The strict transform $D_1 = f_{1*}^{-1}D$ is smooth and meets E_1 at one point p_1 with intersection multiplicity two.

Blow up p_1 to obtain $f_2 : Y_2 \longrightarrow Y_1$, with exceptional divisor $E_2 \cong \mathbb{P}^1$. The smooth curves E_2 , $D_2 = f_{2*}^{-1}D_1$ and $E'_1 = f_{2*}^{-1}E_1$ meet at one point p_2 with three distinct tangent directions. Thus their union has multiplicity three at p_2 , while each pair has intersection multiplicity one.

Blow up p_2 to obtain $f_3 : Y \longrightarrow Y_2$, with exceptional divisor $E_3 \cong \mathbb{P}^1$. The union of E_3 , $D_3 = f_{3*}^{-1}D_2$, $E''_1 = f_{3*}^{-1}E'_1$ and $E'_2 = f_{3*}^{-1}E_2$ is snc. The composition $f = f_1 \circ f_2 \circ f_3$ is a log resolution. We have

$$K_Y + dD_3 = f^*(K_X + dD) + \sum_i a(E_i, X, dD)E_i,$$

where

$$K_Y = f^*K_X + E''_1 + 2E'_2 + 4E_3, \quad f^*D = D_3 + 2E''_1 + 3E'_2 + 6E_3.$$

Therefore

$$\sum_i a(E_i, X, dD)E_i = (1 - 2d)E''_1 + (2 - 3d)E'_2 + (4 - 6d)E_3.$$

Testing these coefficients and the coefficient d of D_3 gives

$$(X, dD) \text{ is klt} \iff 0 \leq d < \frac{5}{6}, \quad (X, dD) \text{ is lc} \iff 0 \leq d \leq \frac{5}{6}.$$

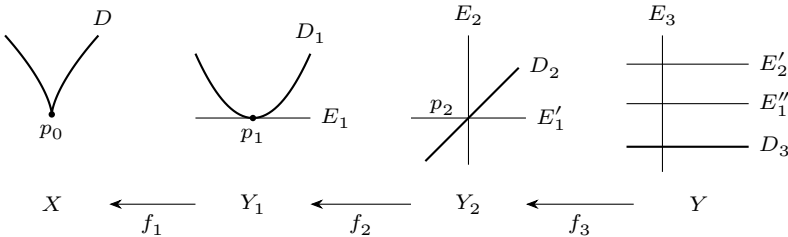


Figure 5.7: Log resolution of (X, dD) .

Lemma 5.2.11 and Corollary 5.2.21 imply the following continuity properties.

Corollary 5.2.38. *Let (X, Δ) be a pair and Δ' an effective $\mathbb{R}$ -Cartier divisor. Then:*

- (1) *If $(X, \Delta + \Delta')$ is terminal (resp. canonical, klt, plt, lc), then (X, Δ) has the same property. The same holds for dlt when $\Delta \geq 0$.*
- (2) *If (X, Δ) is terminal (resp. klt), then $(X, \Delta + \varepsilon \Delta')$ is terminal (resp. klt) for $0 \leq \varepsilon \ll 1$.*
- (3) *If (X, Δ) is plt and Δ, Δ' have no common irreducible components, then $(X, \Delta + \varepsilon \Delta')$ is plt for $0 \leq \varepsilon \ll 1$.*
- (4) *If $\Delta \geq 0$, (X, Δ) is dlt, and*

$$\text{Supp } \Delta' \subseteq \text{Supp}(\Delta - \lfloor \Delta \rfloor),$$

then $(X, \Delta + \varepsilon \Delta')$ is dlt for $0 \leq \varepsilon \ll 1$.

- (5) *If (X, Δ) is terminal, then $(X, \Delta + \Delta')$ is canonical if and only if $(X, \Delta + c\Delta')$ is terminal for every $c < 1$.*
- (6) *If (X, Δ) is klt (resp. plt), then $(X, \Delta + \Delta')$ is lc if and only if $(X, \Delta + c\Delta')$ is klt (resp. plt) for every $c < 1$.*

For $c < 0$ in (5)–(6), we use the convention of Definition 5.2.22 allowing negative boundary coefficients. If “pair” is reserved for effective boundaries, the equivalences can instead be stated for $0 \leq c < 1$.

Proof. We do not assume Δ effective except in the dlt assertions.

For (1)

For every divisor E over X , monotonicity gives

$$a(E, X, \Delta + \Delta') \leq a(E, X, \Delta).$$

This proves the exceptional-discrepancy inequalities, and decreasing the boundary also preserves the coefficient condition for klt. For dlt with $\Delta \geq 0$, keep the closed set Z from Definition 5.2.23. The support of Δ on $X \setminus Z$ is a union of components of the original snc boundary, and the required discrepancies over Z stay greater than -1 .

For (2)

Choose a log resolution $f : Y \rightarrow X$ of $\text{Supp } \Delta \cup \text{Supp } \Delta'$ on which all the strict transforms of its components are pairwise disjoint. Write

$$\Delta = \sum_j d_j D_j, \quad \Delta' = \sum_j e_j D_j,$$

using the same index set and allowing zero coefficients. For the finitely many exceptional divisors E_i , set

$$q_i = a(E_i, X, \Delta), \quad b_i = \text{mult}_{E_i} f^* \Delta' \geq 0.$$

Then

$$a(E_i, X, \Delta + \varepsilon \Delta') = q_i - \varepsilon b_i, \quad \text{coeff}_{D_j}(\Delta + \varepsilon \Delta') = d_j + \varepsilon e_j.$$

Let $t = 0$ in the terminal case and $t = -1$ in the klt case. The initial assumptions give $q_i > t$ and $d_j < 1$. Choose $\varepsilon_0 > 0$ so that

$$\varepsilon_0 \leq \min \left\{ 1, \min_{b_i > 0} \frac{q_i - t}{b_i}, \min_{e_j > 0} \frac{1 - d_j}{e_j} \right\}.$$

Empty minima are omitted. For $0 \leq \varepsilon < \varepsilon_0$, every exceptional discrepancy on this resolution is greater than t , and every boundary coefficient is less than one. Corollary 5.2.21 and the klt coefficient condition prove the assertion. In particular, the klt numerator is $q_i + 1$, rather than q_i .

For (3)

Use the same common resolution. Here $q_i > -1$ and $d_j \leq 1$. The absence of common components means that $e_j > 0$ implies $d_j = 0$. Thus the exceptional inequalities remain strict and no coefficient exceeds one for sufficiently small ε . Corollary 5.2.21 gives discrepancy greater than -1 , so the perturbed pair is plt.

For (4)

Choose a log resolution as in the alternative dlt definition, so that its finitely many exceptional discrepancies are greater than -1 . Since $\text{Supp } \Delta' \subseteq \text{Supp } \Delta$, this also resolves the support of Δ' . The further condition in (4) says that only components whose coefficients are strictly below one are increased. A sufficiently small common $\varepsilon_0 > 0$ therefore preserves all coefficient bounds and all exceptional inequalities on this chosen resolution. The support is unchanged and remains snc upstairs. Definition 5.2.35 shows that the perturbed pair is dlt. The fixed finite resolution is what supplies one value of ε_0 valid for all valuations.

For (5)

For every divisor E and $0 \leq c \leq 1$, discrepancy is affine in c :

$$a(E, X, \Delta + c\Delta') = (1 - c)a(E, X, \Delta) + c a(E, X, \Delta + \Delta').$$

If the endpoint pair is canonical and the initial discrepancy is $\delta > 0$, then every exceptional discrepancy at $c < 1$ is at least $(1 - c)\delta > 0$. Hence the intermediate pair is terminal. Conversely, if every intermediate pair is terminal, taking $c \rightarrow 1$ in the inequality for each fixed E gives $a(E, X, \Delta + \Delta') \geq 0$ for all exceptional E . This is canonicity.

For (6)

Use the same interpolation. If the endpoint is lc and the initial exceptional discrepancy is $\delta > -1$, then for $c < 1$ every exceptional discrepancy is at least

$$(1 - c)\delta - c = -1 + (1 - c)(\delta + 1) > -1.$$

For the klt assertion, each initial boundary coefficient is less than one and each endpoint coefficient is at most one, so its affine interpolation is strictly less than one. For plt, both endpoint coefficient bounds are at most one. This proves the forward direction. Conversely, pass to the limit for each exceptional divisor to get discrepancy at least -1 at the endpoint. In both (5) and (6), negative c cause no further difficulty: monotonicity compares the pair with the initial pair at $c = 0$. $\square$

Proposition 5.2.39. *Let (X, Δ) be a klt pair with $K_X + \Delta$ $\mathbb{R}$ -Cartier.*

(1) *There is a log resolution $f : X' \rightarrow X$ such that, writing*

$$f^*(K_X + \Delta) = K_{X'} + A_{X'} - B_{X'},$$

where $A_{X'}, B_{X'}$ are effective with no common components, the support of $A_{X'}$ is smooth.

(2) *If $a(E, X, \Delta) < 1 + \text{totaldiscrep}(X, \Delta)$, then $\text{center}_{X'} E$ is a divisor. In particular, only finitely many exceptional divisors over X have discrepancy strictly less than $1 + \text{totaldiscrep}(X, \Delta)$.*

Proof. **For (1)**

Start with a log resolution $f : Y \rightarrow X$ and write

$$f^*(K_X + \Delta) = K_Y + A_Y - B_Y, \quad A_Y = \sum_{i \in I} (1 - \alpha_i) A_i,$$

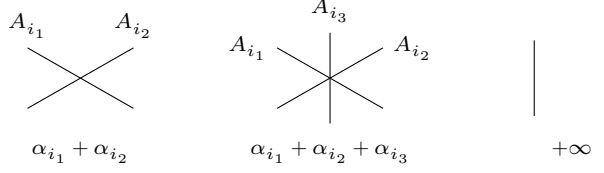
where A_Y, B_Y are effective with no common components and $0 < \alpha_i < 1$. These inequalities follow from klt, since

$$f^*(K_X + \Delta) = K_Y + f_*^{-1} \Delta - \sum_{E_i \text{ exceptional}} a(E_i, X, \Delta) E_i.$$

Let $S(Y) = \sum_{i \in I} \mathbb{Z}_{\geq 0} \alpha_i \subseteq \mathbb{R}_{\geq 0}$. For $J \subseteq I$, set

$$w(Y, J) = \begin{cases} \sum_{i \in J} \alpha_i, & \text{if } \bigcap_{i \in J} A_i \neq \emptyset \text{ and } \#J \geq 2, \\ +\infty, & \text{otherwise.} \end{cases}$$

This records intersections among the components of the positive part:



The middle picture denotes a triple stratum in dimension at least three.

Set

$$M(Y) = \min_{J \subseteq I} w(Y, J), \quad r(Y) = \#\{J \subseteq I : w(Y, J) = M(Y)\}.$$

Either $M(Y) \in (0, \dim X) \cap S(Y)$ or $M(Y) = +\infty$. The latter means that the components of A_Y are pairwise disjoint, i.e. its support is smooth.

Suppose that $M(Y) < +\infty$. Choose J with $w(Y, J) = M(Y)$ and blow up

$$Z = \bigcap_{i \in J} A_i.$$

This is a disjoint union of smooth subvarieties of codimension $c = \#J \geq 2$. Write $p : Y' \rightarrow Y$ for the blow-up and E for its exceptional divisor; the following calculation applies to each component of E . We have

$$K_{Y'} = p^*K_Y + (c-1)E, \quad p^*A_Y = \sum_{i \in I} (1 - \alpha_i) p_*^{-1} A_i + \sum_{i \in J} (1 - \alpha_i) E.$$

No component of B_Y contains a component of Z : the union of the supports is snc, and another component cannot contain a whole stratum of these A_i . Thus p^*B_Y has no exceptional term along E . The coefficient of E in the crepant boundary on Y' is consequently

$$1 - c + \sum_{i \in J} (1 - \alpha_i) = 1 - \sum_{i \in J} \alpha_i = 1 - M(Y).$$

The new positive part is

$$A_{Y'} = \sum_{i \in I} (1 - \alpha_i) p_*^{-1} A_i + \begin{cases} (1 - M(Y))E, & M(Y) < 1, \\ 0, & M(Y) \geq 1. \end{cases}$$

If a new positive component occurs, its weight is $M(Y)$, a sum of old weights. All old components persist, so $S(Y') = S(Y)$. Any intersection involving a new positive component and another positive component has weight strictly greater than $M(Y)$. Old intersections can disappear but cannot be created by the blow-up, and the chosen intersection indexed by J disappears. Hence

$$M(Y') > M(Y), \quad \text{or} \quad M(Y') = M(Y) \text{ and } r(Y') < r(Y).$$

The semigroup generated by finitely many positive real numbers has only finitely many elements in a bounded interval. Thus $M(Y)$ has only finitely many possible finite values. At each fixed value of M , the non-negative integer r strictly decreases. The process must terminate with $M = +\infty$, proving (1). This does not require r to have a fixed bound throughout the whole process.

For (2)

Crepanant pullback and monotonicity give

$$a(E, X, \Delta) = a(E, X', A_{X'} - B_{X'}) \geq a(E, X', A_{X'}). \quad (5.6)$$

If E is exceptional over X' , the smooth support of $A_{X'} = \sum_i (1 - \alpha_i) A_i$ gives

$$a(E, X', A_{X'}) \geq \text{discrep}(X', A_{X'}) = \min_i \{\min \alpha_i, 1\}. \quad (5.7)$$

When $A_{X'} \neq 0$,

$$\text{totaldiscrep}(X, \Delta) \leq \min_i a(A_i, X, \Delta) = \min_i (\alpha_i - 1),$$

so $\min_i \alpha_i \geq 1 + \text{totaldiscrep}(X, \Delta)$. Also $\text{totaldiscrep}(X, \Delta) \leq 0$, since a prime divisor on X outside $\text{Supp } \Delta$ has discrepancy zero. This handles the extra term 1 and the case $A_{X'} = 0$. Thus every E exceptional over X' satisfies

$$a(E, X, \Delta) \geq 1 + \text{totaldiscrep}(X, \Delta).$$

If its discrepancy is *strictly less* than this bound, its center on X' must therefore be a divisor. Only finitely many prime divisors on X' are exceptional over X , proving (2). $\square$

Remark 5.2.40. For any pair with X normal and $K_X + \Delta$ $\mathbb{R}$ -Cartier,

$$\text{totaldiscrep}(X, \Delta) \leq 0.$$

Indeed, there is a prime divisor D outside the finite support of Δ , and $a(D, X, \Delta) = -\text{coeff}_D \Delta = 0$.

Proposition 5.2.41. *Let X be a normal variety and $\Delta = \sum_i a_i D_i$, with distinct prime divisors D_i and $0 \leq a_i \leq 1$. Assume that $K_X + \Delta$ is $\mathbb{R}$ -Cartier. Then (X, Δ) is dlt if and only if there is a closed subset $Z \subseteq X$ such that*

- (1) $X \setminus Z$ is smooth and $\Delta|_{X \setminus Z}$ has snc support;
- (2) there is a log resolution $f : Y \rightarrow X$, also resolving the ideal of Z , such that $f^{-1}(Z)$ is a divisor with snc support together with the transformed boundary, and

$$a(E, X, \Delta) > -1$$

for every prime divisor $E \subseteq f^{-1}(Z)$.

In particular, the inverse image in (2) has pure codimension one when non-empty.

Proof. If the pair is dlt, choose a proper closed bad locus Z as in Definition 5.2.23, and take a log resolution which also principalizes its ideal. Such a Z may be chosen proper because the pair is snc at the generic points of X and of its boundary components. Every component of $f^{-1}(Z)$ has center contained in Z , so the discrepancy inequalities follow from the definition.

Conversely, assume that f exists as in (2). Write

$$K_Y + \Delta_Y = f^*(K_X + \Delta)$$

and choose an effective divisor Δ' supported exactly on $f^{-1}(Z)$. Its components have coefficient less than one in Δ_Y . Every other component of Δ_Y has coefficient at most one: those which meet $f^{-1}(X \setminus Z)$ are governed by the lc snc pair there. For sufficiently small $\varepsilon > 0$, the divisor $\Delta_Y + \varepsilon\Delta'$ therefore has snc support and all coefficients at most one. It is lc by Corollary 5.2.20.

For any divisor E with $\text{center}_X E \subseteq Z$, properness gives $\text{center}_Y E \subseteq f^{-1}(Z)$. Thus

$$a(E, X, \Delta) = a(E, Y, \Delta_Y) > a(E, Y, \Delta_Y + \varepsilon\Delta') \geq -1.$$

This proves Definition 5.2.23(2). $\square$

Proposition 5.2.42. *For an effective $\mathbb{R}$ -divisor Δ , the pair (X, Δ) is klt if and only if it is dlt and $\lfloor \Delta \rfloor = 0$.*

Proof. For the forward direction, all divisorial discrepancies of a klt pair are greater than -1 , and all boundary coefficients are less than one. In Definition 5.2.23 we may take $Z = X$, so the pair is dlt.

Conversely, suppose that the pair is dlt and $\lfloor \Delta \rfloor = 0$, and let Z be its bad locus. If $\text{center}_X E \subseteq Z$, then $a(E, X, \Delta) > -1$ by definition. Otherwise restrict to $U = X \setminus Z$. The snc formula gives

$$\text{discrep}(U, \Delta|_U) = \min \left\{ \min_{i \neq j, D_i|_U \cap D_j|_U \neq \emptyset} (1 - a_i - a_j), \min_i (1 - a_i), 1 \right\} > -1,$$

since every $a_i < 1$. This equals the corresponding bound for valuations whose centers meet U . All exceptional discrepancies are thus greater than -1 ; the finite-resolution formula supplies a uniform strict lower bound. Together with $\lfloor \Delta \rfloor = 0$, this proves klt. $\square$

Recall that for a Weil divisor D , its rank-one reflexive sheaf $\mathcal{O}_X(D)$ is coherent and

$$H^0(X, \mathcal{O}_X(D)) = \{h \in \mathbb{C}(X)^\times : \text{div}(h) + D \geq 0\} \cup \{0\}.$$

We can define its complete linear series by

$$\begin{aligned} |D| &= \{D_h : D_h \text{ is effective and } D_h \sim_{\text{lin}} D\} \\ &= \{\text{div}(h) + D : 0 \neq h \in H^0(X, \mathcal{O}_X(D))\}, \end{aligned}$$

and its set-theoretic base locus by

$$\text{Bs}(|D|) = \bigcap_{D_h \in |D|} \text{Supp } D_h.$$

For a finite-dimensional subspace $V \subseteq H^0(X, \mathcal{O}_X(D))$, there is a map

$$\Phi : \mathbb{P}(V) \longrightarrow |D|, \quad [h] \longmapsto \text{div}(h) + D.$$

Here h is the rational function representing the section. In the earlier section notation, the same divisor is simply $\text{div}(s)$; one must not add D twice. The map is well-defined, but on a non-proper variety it need not be injective, since non-constant units can give the same divisor. Its image defines the corresponding linear subseries. If X is proper and integral over $\mathbb{C}$, its global units are constant and this map is injective.

Recall that $D_1 \sim_{\mathbb{R}} D_2$ (resp. $D_1 \sim_{\mathbb{Q}} D_2$) means that $D_1 - D_2$ is a real (resp. rational) linear combination of principal Weil divisors.

A dlt pair can be approximated by klt pairs as follows:

Proposition 5.2.43. *Assume that (X, Δ) is dlt and X is quasi-projective with an ample Cartier divisor H . Let Δ_1 be an effective $\mathbb{R}$ -divisor, not necessarily $\mathbb{R}$ -Cartier, such that $\Delta - \Delta_1 \geq 0$. Then there are a real number $c > 0$ and an effective $\mathbb{R}$ -divisor $D \sim_{\mathbb{R}} \Delta_1 + cH$ such that*

$$(X, \Delta - \varepsilon \Delta_1 + \varepsilon D)$$

is dlt for all $0 < \varepsilon \ll 1$. If $\text{Supp } \Delta_1 = \text{Supp } \Delta$, these pairs are klt for all sufficiently small $\varepsilon > 0$.

Before proving this result, we need a rational version.

Lemma 5.2.44. *Assume that (X, Δ) is dlt and X is quasi-projective with an ample Cartier divisor H . Let Θ be an effective $\mathbb{Q}$ -divisor, not necessarily $\mathbb{Q}$ -Cartier, such that $\Delta - \Theta \geq 0$. Then there are a rational number $c > 0$ and an effective $\mathbb{Q}$ -divisor $B \sim_{\mathbb{Q}} \Theta + cH$ such that*

$$(X, \Delta - \varepsilon \Theta + \varepsilon B)$$

is dlt for all $0 < \varepsilon \ll 1$. If $\text{Supp } \Theta = \text{Supp } \Delta$, these pairs are klt for all sufficiently small $\varepsilon > 0$.

Proof. We have $\Delta = (\Delta - \Theta) + \Theta$ with both summands effective.

Let $Z \subseteq X$ be a closed subset witnessing that (X, Δ) is dlt, and put $U = X \setminus Z$. Choose positive integers m, m' such that $m\Theta$ is integral and the coherent sheaf

$$\mathcal{O}_X(m\Theta + m'H)$$

is generated by global sections. This uses ampleness of H and does not require $m\Theta$ to be Cartier. Since X is quasi-projective, the whole space of global sections need not be finite-dimensional. Nevertheless, quasi-compactness and coherence let us select finitely many global sections which generate the sheaf: on each of finitely many affine opens, finitely many of the given global generators suffice. Let V be their finite-dimensional span.

On the smooth open set U , the sheaf is a line bundle and the induced system is basepoint-free. Apply Bertini both on U and on all strata of $\text{Supp}(\Delta|_U)$. For a general member, its effective divisor

$$A \sim_{\text{lin}} m\Theta + m'H$$

is smooth on U and meets the boundary strata transversely, so $A|_U + \Delta|_U$ has snc support. We also choose A to contain no component of Δ . This is possible because the sheaf is generated at the generic point of each of those components. Applying Bertini on X_{reg} shows that A is reduced in codimension one, hence is a reduced Weil divisor.

Set $B = A/m$ and $c = m'/m$. Then B is effective and

$$B \sim_{\mathbb{Q}} \Theta + cH.$$

In particular, $B - \Theta$ is $\mathbb{Q}$ -Cartier, although B and Θ need not be individually $\mathbb{Q}$ -Cartier. Therefore

$$K_X + \Delta - \varepsilon\Theta + \varepsilon B = K_X + \Delta + \varepsilon(B - \Theta)$$

is $\mathbb{R}$ -Cartier for every real ε .

Choose a log resolution $f : Y \rightarrow X$ of $(X, \Delta + A)$ which is an isomorphism over the snc open set U . Every exceptional prime divisor E on this resolution has center in Z , so $a(E, X, \Delta) > -1$. We have

$$a(E, X, \Delta - \varepsilon\Theta + \varepsilon B) = a(E, X, \Delta) - \varepsilon \text{ord}_E f^*(B - \Theta).$$

There are only finitely many such E . Hence one common $\varepsilon_0 > 0$ keeps all these numbers greater than -1 . Decrease ε_0 further so that $\varepsilon_0 \leq \min\{1, m\}$. Then $\Delta - \varepsilon\Theta$ is effective with coefficients at most one, and the new components of εB have coefficient $\varepsilon/m < 1$. Since A contains no component of Δ , the entire boundary has coefficients at most one. Its total transform is supported in the fixed snc divisor on Y . The alternative dlt definition now proves the first assertion.

For the last assertion, write $\Delta = \sum_i a_i D_i$ and $\Theta = \sum_i b_i D_i$, with the same support and $b_i > 0$. We may further require

$$\varepsilon_0 \leq \min \left\{ m, \min_i \frac{a_i}{b_i} \right\},$$

omitting the minimum if the support is empty. For $0 < \varepsilon < \varepsilon_0$,

$$0 \leq a_i - \varepsilon b_i < 1,$$

and the new components from B have coefficients less than one. Therefore

$$\lfloor \Delta - \varepsilon \Theta + \varepsilon B \rfloor = 0.$$

By Proposition 5.2.42, these dlt pairs are klt. □

We now prove Proposition 5.2.43.

Proof of Proposition 5.2.43. Our strategy is to use Lemma 5.2.44 and rational approximation. Write

$$\Delta_1 = \sum_i r_i \Gamma_i, \quad r_i > 0.$$

Choose rational numbers $0 < q_i < r_i$, and set $\Theta = \sum_i q_i \Gamma_i$. If $\Delta_1 = 0$, take $\Theta = 0$. Then Θ is effective with $\text{Supp } \Theta = \text{Supp } \Delta_1$, and

$$\Delta - \Theta = (\Delta - \Delta_1) + (\Delta_1 - \Theta) \geq 0.$$

Apply Lemma 5.2.44 to get $c \in \mathbb{Q}_{>0}$ and an effective $\mathbb{Q}$ -divisor B with

$$B \sim_{\mathbb{Q}} \Theta + cH, \tag{5.8}$$

such that $(X, \Delta - \varepsilon \Theta + \varepsilon B)$ is dlt for $0 < \varepsilon \ll 1$.

Set $D = B + (\Delta_1 - \Theta)$, which is effective. By (5.8),

$$D - (\Delta_1 + cH) = B + (\Delta_1 - \Theta) - \Delta_1 - cH = B - \Theta - cH$$

is a rational linear combination of principal divisors, hence also a real one. Thus $D \sim_{\mathbb{R}} \Delta_1 + cH$. Finally,

$$\begin{aligned} \Delta - \varepsilon \Delta_1 + \varepsilon D &= \Delta - \varepsilon \Delta_1 + \varepsilon (B + \Delta_1 - \Theta) \\ &= \Delta - \varepsilon \Theta + \varepsilon B. \end{aligned}$$

The first assertion follows. If $\text{Supp } \Delta_1 = \text{Supp } \Delta$, then $\text{Supp } \Theta = \text{Supp } \Delta$, so the last assertion of Lemma 5.2.44 gives klt as well. □

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